A Unified Generative Adversarial Network With Convolution and Transformer for Remote Sensing Image Fusion

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***Abstract*— Images derived from an individual sensor fail to simultaneously satisfy the demands of high spatial, spectral, and temporal resolutions. Multisource remote sensing image (RSI) fusion provides efficient access to high-spatial-resolution multispectral (HRMS) images [spatial–spectral fusion (SSF)] and high temporal- and spatial-resolution images [spatiotemporal fusion (STF)]. While existing deep learning (DL)-based models can mainly implement either SSF or STF, there is an urgent need for models that can simultaneously implement both SSF and STF. A unified generative adversarial network with convolution and Transformer (CTUGAN) for SSF and STF is proposed. CTUGAN contains a adaptive convolutional Transformer generator (ACTG) and multiresolution convolutional Transformer discriminator (MCTD), both with the convolution and Transformer. First, a bidirectional local–global feature encoder is devised in the ACTG to extract local–global features via a high-to-low resolution and a low-to-high resolution. Then, a multihead cross-attention fusion decoder (MCAFD) is devised to aggregate and fuse complementary local–global features of various levels and resolutions hierarchically to restore valuable information. Moreover, MCTDs adversely learn multiresolution local–global features to identify the relative reality of products, and a generalized loss function is built to accomplish full supervision. Finally, numerous experiments on the SSF data (Gaofen-2 (GF-2) and QuikBird) and STF data [Coleambally Irrigation Area (CIA) and lower Gwydir catchment (LGC)] demonstrate that the proposed CTUGAN model outperforms both subjective and objective evaluations.**

***Index Terms*— Bidirectional local–global feature encoder, convolution and Transformer, multihead cross-attention fusion, multiresolution convolutional Transformer discriminators, remote sensing image (RSI) unified fusion model.**

I. Introduction

REMOTE sensing images (RSIs) with high temporal and spatial resolution play remarkably significant roles in the monitoring of ecological environment changes, marine resources, and dynamic urban planning [1], [2]. Nevertheless, products are mutually restricted in temporal, spatial, and spectral resolutions caused by constraints of the sensor technology and other factors, which largely limits the application of RSIs [3], [4]. The temporal and spatial resolutions are mutually constrained, for example, the revisit period of the Landsat satellite is about 16 days, and the spatial resolution of the multispectral (MS) image is 30 m; whereas the period of MODIS image is one day and the spatial resolution is 500 m [5], [6]. The spectral and temporal resolutions are mutually limited, for instance, the spatial resolution of the MS image with four bands captured by the Gaofen-2 (GF-2) satellite is about 4 m, whereas the panchromatic (PAN) image is one band and has a spatial resolution of 1 m [7], [8]. Multisource remote sensing fusion provides a flexible and effective scheme for producing high-resolution images by incorporating complementary information derived from multiple sources. Presently, numerous RSI fusion methodologies have been developed, which are categorized into spatial–spectral fusion (SSF), spatiotemporal fusion (STF), and spatiotemporal–spectral fusion (STSF) [9].

A. Spatial–Spectral Fusion

SSF aggregates complementary spatial and spectral information between multisource RSIs to yield the high-spatial-resolution MS (HRMS) image [8]. According to the image modalities, SSF involves PAN and MS image fusion, PAN and hyperspectral image fusion, and MS and hyperspectral image fusion [7], [8]. We mainly focus on PAN and MS image fusion.

The SSF models are roughly categorized into component substitution (CS), multiresolution analysis (MRA), variational optimization (VO), and deep learning (DL) [7]. The CS models enhance the spatial resolution of MS images, and however, oversharpening and severe spectral distortion can arise. Typical approaches of the CS are the Gram–Schmidt (GS) transformation and GS adaptive (GSA) [10], principal component analysis [11], band-dependent spatial detail (BDSD) injection [12], and so forth.

Compared with the CS approach, the MRA methods decrease the oversharpening and spectral distortion, which cause detail loss and relative blurring. Representative methodologies are smoothing-filter-based intensity modulation (SFIM) [13], modulation transfer function generalized Laplacian pyramid transform (MTF-GLP) [14],

Manuscript received 3 April 2024; revised 21 June 2024; accepted 10 July 2024. Date of publication 12 August 2024; date of current version 23 August 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 82260362 and in part by Hainan Major Science and Technology Program of China under Grant ZDKJ202017. *(Corresponding author: Mengxing Huang.)*

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Digital Object Identifier 10.1109/TGRS.2024.3441719

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MTF-GLP-high-pass modulation (MTF-GLP-HPM) [15], MTF-GLP-context-based decision [16], and so on.

The VO-based method treats PAN sharpening (pansharpening) as an optimization problem, which can be constructed using energy functions and optimization methods. Bayesian-based methods [17] and sparse representation-based methods can also be categorized into this group, such as convolutional sparse representation of injected detail for pansharpening [18] and gradient sparse representation-based pansharpening [19]. Simoes et al. [20] treated hyperspectral and MS image fusion as a convex function optimization problem, which estimates both spatial blur and spectrum linear operators to implement effective fusion. VO-based methodologies also involve variational pansharpening with cartoon-texture similarity [21], variational pansharpening with spectral gradient fidelity and spatial Hessian hyper-Laplacian sparsity constraints [22], nonlocal regression coefficient estimation [23], a pansharpening model via learnable nonlinear mapping [24], and so forth. VO-based approaches provide superior spectral fidelity compared to CS/MRA-based approaches. However, the assumptions of most methods may not completely match the real-life scenario, which causes suboptimal performance. Moreover, the VO-based models are complex, and the calculation is more complicated.

DL-based schemes are summarized as supervised SSF and unsupervised SSF. For supervised SSF models, Cao et al. [25] realized hyperspectral image super-resolution using window-based linear mixing and convolutional neural network-based (CNN) feature transfer to leverage spectra and multilevel features. Shao et al. [26] devised a dual-stream residual encoder–decoder conditional generative adversarial network (RED-cGAN), employing the PAN image as a condition to boost details. Liu et al. [27] presented a pansharpening GAN (PsGAN), which comprises a dual-stream convolutional generator and a convolutional discriminator for adversarial learning to maintain identical distribution between the production and the reference data. Benzenati et al. [28] devised a two-stream detail injection GAN (DI-GAN), which predicts high-frequency information and utilizes a relative discriminator to render the production more realistic. Meng et al. [29] proposed a ViT-based SSF model, which utilizes a Transformer encoder to establish long-distance dependencies and produce HRMS images. Zhou et al. [30] built a two-stream Transformer model for pansharpening (PanFormer), which exploits the self-attention to extract MS and PAN features, the cross-attention to fuse spectral and spatial information, and convolutions to reconstruct the HRMS. Hou et al. [31] introduced a two-stream PAN-oriented multiresolution fusion network, which designs a Swin Transformer-based residual attention module to enhance the feature representation.

Targeting the absence of reference images, Ma et al. [32] introduced an unsupervised pansharpening model (PanGAN), which combines spectral and spatial discriminators to preserve spectral and spatial information to generate high-quality images. Zhou et al. [33] implemented an unsupervised GAN (PGMAN) and an unsupervised cycle-consistent GAN (UCGAN) [34] to promote image quality. To promote the TDPNet [35], Wu et al. [36] proposed an unsupervised GAN with recursive mixed-scale feature fusion to boost the image quality. Li et al. [37] realized a no-reference pansharpening incorporating CNN and pyramid Transformer (CPT-noRef), which extracts shallow local and multiscale global features, respectively, and utilizes no-reference spatial and spectral losses to optimize the model. Xiong et al. [38] built an unsupervised attention pansharpening network (UAP-Net), which augments details by spatial texture attention and devises dual-domain spatial and spectral losses. Cao et al. [39] presented an unsupervised model with CNN and Transformer to fulfill hyperspectral and MS fusion, which utilizes CNN to implement two reduced-resolution networks and multilevel cross-attention to aggregate hyperspectral and MS features.

CNN-based SSF models have achieved some results, and in recent years, the Transformer has been gradually adopted in SSF. While most models are trained on the reduced-resolution data, though good results are attained on the reduced-resolution data, the results on the full-resolution data are not favorable, making it undesirable for real-world applications. The SSF has considerable room to advance in the spectra and details retention for the full-resolution outcome. The GAN- and Transformer-based unsupervised models also have huge potential to enhance the spectra and details retention.

B. Spatiotemporal Fusion

STF aggregates a sparse-time high-spatial-resolution image and a dense-time low-spatial-resolution image to yield a high-temporal and high-spatial-resolution (HTHR) image. The STF techniques roughly cover weighting-based, unmixing-based, hybrid, and learning-based models [5].

For traditional approaches, Gao et al. [40] presented a spatiotemporal adaptive reflectance fusion model (STARFM), which utilizes prior Landsat–MODIS data and MODIS data at the prediction date (three images) to produce the Landsat data at the prediction date. STARFM is unsuitable for complex heterogeneous scenarios. Zhu et al. [41] introduced an enhanced STARFM (ESTARFM), which produces the Landsat image using two pairs of prior Landsat–MODIS images and the MODIS image of the predicted date (five images) to enhance precision. However, the ESTARFM has more challenges in forecasting scenarios with larger abrupt changes. Zhu et al. [42] proposed a flexible spatiotemporal data fusion (FSDAF) approach based on spectrum unmixing and thin-plate spline interpolation for forecasting gradual changes and land-cover-type changes, which works for heterogeneous landscapes.

Regarding DL-based models, Tan et al. [43] presented an enhanced deep convolutional STF network (EDCSTFN), which utilizes CNNs to obtain high-frequency information and variations and devises a composite loss function. Tan et al. [44] introduced a conditional GAN with the switchable normalization for the STF (GAN-STFM), which utilizes two images, overcomes the temporal constraints of the prior data, and improves the application in reality. Liu et al. [45] proposed a progressive spatiotemporal attention fusion GAN (PSTAF-GAN), which acquires multiscale change

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information and high-frequency information at the feature layer to increase image accuracy. Chen et al. [46] devised an STF fusion model based on the Swin Transformer (SwinSTFM), which takes a lot of training data to avoid overfitting. High-quality pairwise data are rare, so data augmentation or a pretrained model is required. Liu et al. [47] developed a detail injection-based STF model, which needs five inputs and devises a three-branch detail injection module, fine detail prediction module, and reconstruction module.

Although the STF techniques perform well in forecasting phenological changes, predicting abrupt land-cover changes remains challenging. Furthermore, the enormous difference in resolution renders the STF difficult to forecast the details of changes; the STF accuracy is greatly affected by the number of reference images and the degree of changes, making it difficult to apply in real scenarios.

C. Spatiotemporal–Spectral Fusion

The aforementioned SSF and STF are independently implemented, so an integrated STSF framework has been developed to implement SSF, STF, and STSF. Shen et al. [48] devised an integrated STSF model to accomplish multiview fusion, SSF, STF, and multisource STSF, which is unconstrained by the number of images and applies to homogeneous images. The performance has a large room for improvement. Zhao et al. [49] developed a sparse representation model that learns spatial–spectral and temporal variation features to achieve SSF, STF, and STSF, which is inapplicable for multisource heterogeneous data and has high costs. Peng et al. [50] built an integrated STSF model based on semicoupled sparse tensor factorization, which implements SSF, STF, and STSF, but the capability of expressing changes is limited. Jiang et al. [51] designed a deep residual CycleGAN, which enhances the spatial resolution and removes clouds via heterogeneous data fusion. Wu et al. [52] developed a multiresolution GAN with bidirectional adaptive-stage progressive oriented fusion (BPF-MGAN), which achieves SSF and STF.

For diverse remote sensing needs, there are different requirements for the spatial, temporal, and spectral resolution of RSIs. HRMS images play a significant role in increasing the accuracy of remote sensing tasks such as natural resources interpretation, land-cover classification, and target recognition, which provides reliable and high-quality data support [7]. HTHR MS images play an important role in improving the reliability of eco-spatial and marine spatial resource monitoring, dynamic change detection, disaster monitoring, and early warning [5].

Currently, the traditional integrated STSF framework is limited by capacity and complex model computation, while the DL model has a strong nonlinear expression capability. However, SSF and STF are developed independently, the vast majority of DL-based models can only realize SSF or STF, and the relationship of spatiotemporal–spectral features applicable to both SSF and STF is seldom researched, which results in the lack of applicability and general functioning of the models. Therefore, focusing on the different application requirements of remote sensing, it is highly meaningful to research the DL-based RSI STSF model to integrate SSF and STF, which produces high-resolution MS images to meet the demands of different tasks, such as RSI information extraction and dynamic change monitoring in ecological space and marine space. The development and mining of the DL-based integrated STSF framework are also essential trends for the advancement of multisource RSI fusion technology. Consequently, compelled by the demand for diverse remote sensing applications, a unified generative adversarial fusion model is proposed to realize both SSF and STF to generate HRMS images and HTHR MS images that satisfy various tasks.

Nevertheless, the implementation of a unified STSF model of multisource RSIs is still encountering the following problems.

1) The existing integrated fusion models have limited capability due to the complicated relationships between multisource RSIs. The DL model provides a robust nonlinear expression capability, but the vast majority of DL-based models can only implement SSF or STF, and very rarely DL-based models enable both SSF and STF to fulfill the demands of various tasks such as natural resource interpretation, target detection and recognition, and dynamic change monitoring. Therefore, the problem of the lack of a DL-based integrated fusion model for SSF and STF needs to be solved.
2) RSIs derived from various sensors with different imaging mechanisms possess various resolutions, and the bands of multisource images are incompletely covered. The significant resolution difference and sensor bias between multisource RSIs cause difficulties in reconstructing textures, details, and changes in low-resolution images. For instance, the ratio of resolution between MS and PAN images in SSF is 4, whereas the ratio of resolution between the MODIS and Landsat images in STF is about 16. The problem of how to overcome the huge resolution gap and sensor differences and sufficiently aggregate the effective information of various source images needs to be solved.
3) The absence of reference data for supervised training, for example, the SSF model, lacks the HRMS image, and the trained reduced-resolution model yields unsatisfactory performance on the full-resolution data, whereas STF has the reference image. Therefore, how to make the integrated model unconstrained by the no-reference image is a challenge to be solved.

Taking the aforementioned challenges into account, based on our previous work BPF-MGAN [52], a unified generative adversarial network with convolution and Transformer (CTUGAN) is proposed to implement SSF and STF. Ablation experiments are executed on the SSF and STF datasets, and the CTUGAN is compared with the prevailing SSF and STF models. The experimental outcomes verify the superior performance of the proposed CTUGAN model in SSF and STF. The main contributions are given as follows.

1) A unified GAN with convolution and Transformer is proposed to construct latent relations of multisource

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data realizing SSF and STF, which contains a adaptive convolutional Transformer generator (ACTG) and multiresolution convolutional Transformer discriminators (MCTDs), to satisfy the demands of high-resolution images for different missions.

2) A bidirectional adaptive local–global feature encoder is devised in the ACTG, which adaptively extracts low-to-high and high-to-low resolutions local–global features, and a residual multihead depthwise convolutional Transformer (RMDCT) is proposed to enhance the representation of multiscale local–global features of RSIs.
3) A multihead cross-attention fusion decoder (MCAFD) is devised in ACTG, which proposes multiresolution feature fusion (MFF) and multihead cross-attention fusion Transformer (MCAFT) to autonomously aggregate complementary local–global information and strengthen the cross-resolution-hierarchy features to recover the effective information.
4) MCTDs are devised to adversely learn the multiresolution local–global structure and spectrum and provide feedback to the ACTG; furthermore, a generalized loss function is devised to form deep hybrid supervision and optimize the model performance.

The remaining sections are allocated as follows. Section II illustrates the proposed model and the generalized loss function. Section III presents datasets, evaluation indicators, SSF and STF comparative experiments, and ablation experiments. Section IV provides discussions. Section V reaches conclusions.

II. Methodology

A. Motivation and Unified Framework

The key notations utilized are denoted as follows: FI denotes the fine-spatial-resolution image of inputs, CI signifies the coarse-spatial-resolution image of inputs, and GI indicates the generated image. F_n and C_n ($n = 0, 1, \ldots, N$) indicate the nth-scale features. The height and width of F_0 are identical to those of FI (maximum spatial-resolution dimension: $H \times W \times d$), and the height and width of F_N are equal to those of CI (minimum spatial-resolution dimension: $h \times w \times B$). B and d are set according to the channel number of inputs for SSF and STF tasks, respectively. The spatial-resolution ratio between F_0 and F_n is τ^n, τ is the downsampling factor, and $\tau = 2$. Y_n refers to the nth-scale intermediate image. HI_n indicates the nth-scale predicted image and HI_0 indicates the final predicted image.

First, facing the complex relations between multisource RSIs, the limited capability of existing integrated fusion schemes, and the fact that rare DL-based models can realize both SSF and STF, a unified GAN with convolution and Transformer is proposed, which can implement both SSF and STF. The integral architecture is illustrated in Fig. 1, which contains an ACTG and MCTDs.

Second, the target sizes in RSIs are diverse, and single-scale features are not sufficient to express information about multisize targets; thus, multiresolution features of fine- and coarse-resolution images are extracted to express information about multisize targets.

Third, RSIs captured by different sensors with different imaging mechanisms exhibit distinct resolutions and incomplete coverage of bands. Due to the huge resolution between the CI and the FI, the information in the CI is limited. The majority of models directly exploit CI features that are upsampled to the dimension of the FI, which increases the huge resolution at a time and introduces much uncertainty to the fusion [53]. Consequently, to decrease the spectral loss and detail distortion that may be caused by improving the large resolution at one time, a smaller factor is considered to gradually enhance the resolution. The spatial structure of the CI is enhanced and extracted step-by-step, and the dual-source information is fused to improve the image resolution step-by-step. Furthermore, depending on the fine- and coarse-resolution characteristics of dual-source images, multiresolution features of the CI are extracted from the coarse-resolution to fine-resolution direction; multiresolution features of the FI are extracted from the fine-resolution to coarse-resolution direction.

Fourth, for images, the potential relationship between local blocks within a large spatial extent of an image or multiple images is termed long-range dependencies. RSIs comprise richer information than natural images, and the pixels of RSI are interconnected. For the multisource RSI fusion task, the long-range dependency involves the latent relationship between CI blocks, the latent relationship between FI blocks, and the latent relationship between CI blocks and FI blocks over a large spatial range. The limited receptive field of convolution in CNNs has the advantage of capturing local features but exhibits difficulties in capturing global information and long-range dependencies between pixels of RSIs. Transformer [54] is an effective encoder–decoder model for language translation. Transformer can establish and utilize long-range dependencies to access any portion of the data through self-attention operations. However, the complexity of the self-attention model presents a quadratic increment with growing spatial resolution. To decrease the computation burden, Liang et al. [55] and Wang et al. [56] applied self-attention by reducing the window size, i.e., 8×8; Chen et al. [57] split the image into nonrepeating blocks and calculated the self-attention in each block independently. However, such operations constrain the spatial extent of self-attention and contradict the goal of capturing true long-range pixel relations, particularly on high-resolution images. To tackle the aforementioned problems, Restormer [58] presented a multi-Dconv head transposed attention block to replace the vanilla multihead self-attention for establishing long-range dependencies, which makes the model complexity linearly growing and is applicable to high-resolution images. Moreover, Uformer [56] also augments the local context by a locally enhanced feedforward module.

Incorporating the above four considerations, a bidirectional adaptive local–global feature encoder (BALGE) and an MCAFD are devised in the ACTG to tackle the difficulty of reconstructing textures, details, and changes information in low-resolution images, which is caused by the huge

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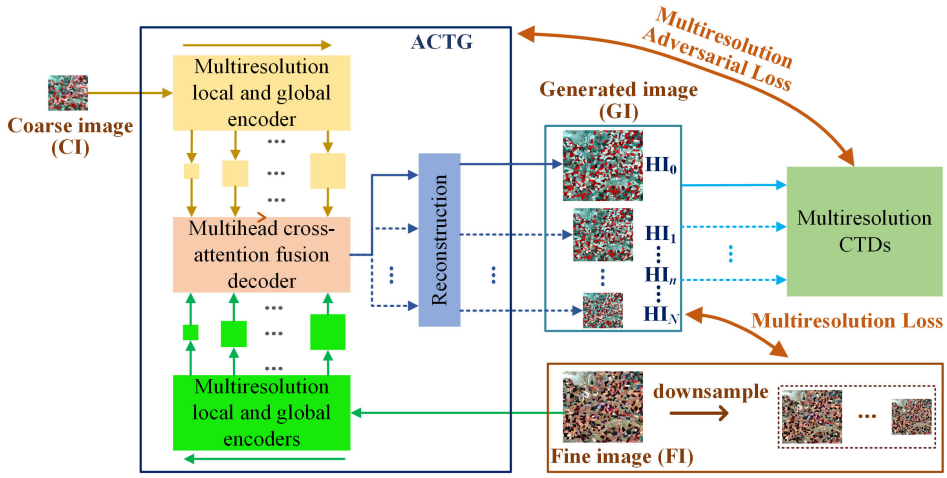


Fig. 1. Integral architecture of a CTUGAN.

resolution difference and sensor bias between multisource RSIs. Moreover, in the BALGE, an RMDCT is proposed by incorporating convolution and Transformer to strengthen the representation of multiscale local and global features of RSIs. RMDCT can benefit from the self-attention in Transformer to establish long-range dependencies of RSI, benefit from convolution to capture the local conception information of input data, and can be applied to high-resolution RSIs. The BALGE can adaptively tune the order N and factor τ according to the resolution of CI and FI, which extracts low-to-high and high-to-low resolutions local–global features. Furthermore, in the MCAFD, MFF and MCAFT are devised to autonomously aggregate and strengthen complementary cross-resolution-hierarchy features for recovering effective local–global information.

Finally, aiming at the lack of reference data for supervised training, MCTDs are devised to adversely learn the multiresolution local–global structure and spectrum and provide feedback to the ACTG; furthermore, a generalized loss function is established to form deep hybrid supervision.

B. Generator

An ACTG is proposed, the architecture is presented in Fig. 2. The ACTG is constructed by the BALGE, MCAFD, and a reconstruction unit (RU). The BALGE consists of two bidirectional branches: one branch hierarchically encodes multiresolution local–global features of the FI image in a high-to-low resolution direction, while the other branch hierarchically encodes multiresolution local–global features of the CI image in a low-to-high resolution direction. MCAFD is devised to integrate and reinforce multisource multiresolution local–global features and restore valuable information. RU is applied to reconstitute information and generate the final predicted result.

1) Bidirectional Adaptive Local–Global Feature Encoder: As illustrated in Fig. 2, the branch for extracting multiresolution local–global features of the FI image consists of a dilated convolution block (MDconv), RMDCTs, and downsampling. The branch for extracting multiresolution local–global features of the CI image consists of an MDconv block, RMDCTs, and upsampling. In the BALGE, RMDCT utilizes dilated convolution to strengthen the multiscale local information and convolutional Transformer to enhance the local–global features for better representation of spectra, textures, and details.

The dilated convolution is utilized to extract shallow multiscale local features of the FI and CI images, as exhibited in Fig. 3(a). First, multiscale features of the FI and CI are extracted in parallel from convolutions with rates of 1, 2, and 3. Then, the multiscale features are concatenated and preliminarily fused using a convolution with a kernel of 1×1. Dconv(k3,r1) means the dilated convolution with a kernel of 3×3 and a rate of 1. The realization process is presented in the following equations:

$$\mathrm{FI}_i = \kappa_{\mathrm{lrelu}}\left(\kappa^{3,i}_{\mathrm{dconv}}(\mathrm{FI})\right) \quad i = 1, 2, 3 \tag{1}$$

$$\mathrm{SF}_0 = \kappa^{1}_{\mathrm{convl}}(\kappa_{\mathrm{concat}}(\mathrm{FI}_1, \mathrm{FI}_2, \mathrm{FI}_3)) \tag{2}$$

$$\mathrm{CI}_i = \kappa_{\mathrm{lrelu}}\left(\kappa^{3,i}_{\mathrm{dconv}}(\mathrm{CI})\right) \quad i = 1, 2, 3 \tag{3}$$

$$\mathrm{SC}_N = \kappa^{1}_{\mathrm{convl}}(\kappa_{\mathrm{concat}}(\mathrm{CI}_1, \mathrm{CI}_2, \mathrm{CI}_3)) \tag{4}$$

where SF_0 and SC_N signify the shallow features of FI and CI, respectively; $\kappa^{3,i}_{\mathrm{dconv}}(\cdot)$ indicates a dilated convolution operation with a kernel size of 3 and a rate of i; and FI_i and CI_i indicate the FI and CI features obtained via the dilated convolution, respectively. $\kappa^{1}_{\mathrm{convl}}(\cdot)$ refers to the convolution with a kernel of 1 followed by a leaky rectified linear unit (LReLU) function [59], $\kappa_{\mathrm{lrelu}}(\cdot)$ means the LReLU function, and $\kappa_{\mathrm{concat}}(\cdot)$ denotes the concatenation operation along the channels.

To capture multiresolution spatial details and spectra of the FI and CI images utilized for fusion, the RMDCT combined with downsampling or upsampling is designed. As depicted in Fig. 2, the $N+1$ resolution local–global features of the FI image are accessed via $N+1$ RMDCTs and N downsampling operators. Likewise, the $N+1$ resolution local–global features of the CI image are acquired via $N+1$ RMDCTs and N upsampling operators. The realization process is presented in (5)–(10). The downsampling and upsampling realized by pixel unshuffle and pixel shuffle are depicted in Fig. 3(b), and

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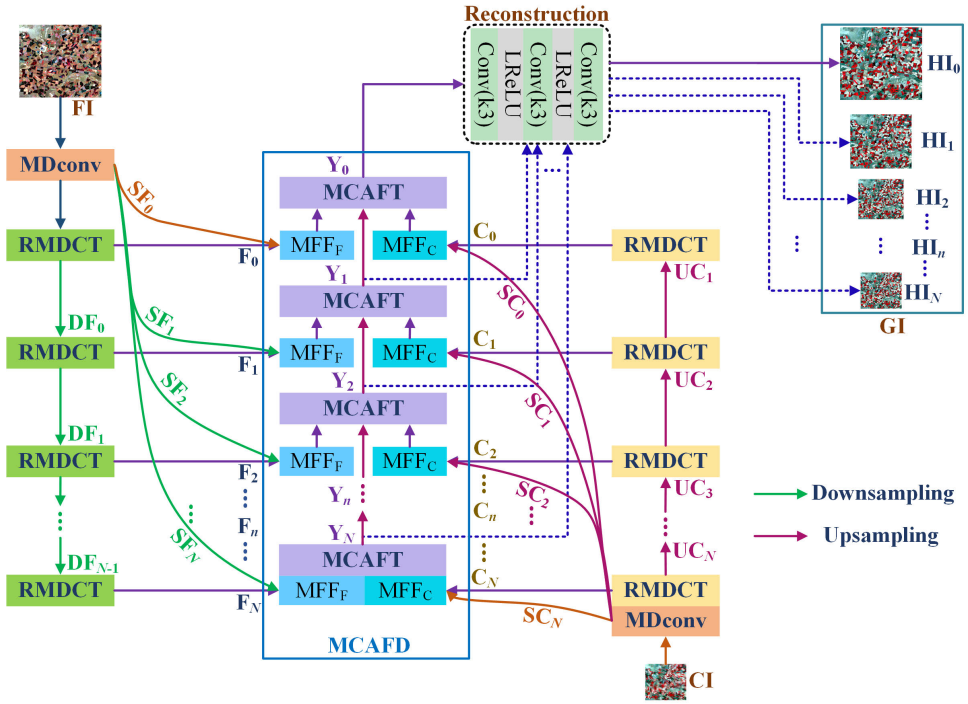


Fig. 2. Architecture of a ACTG with a bidirectional encoder and fusion decoder.

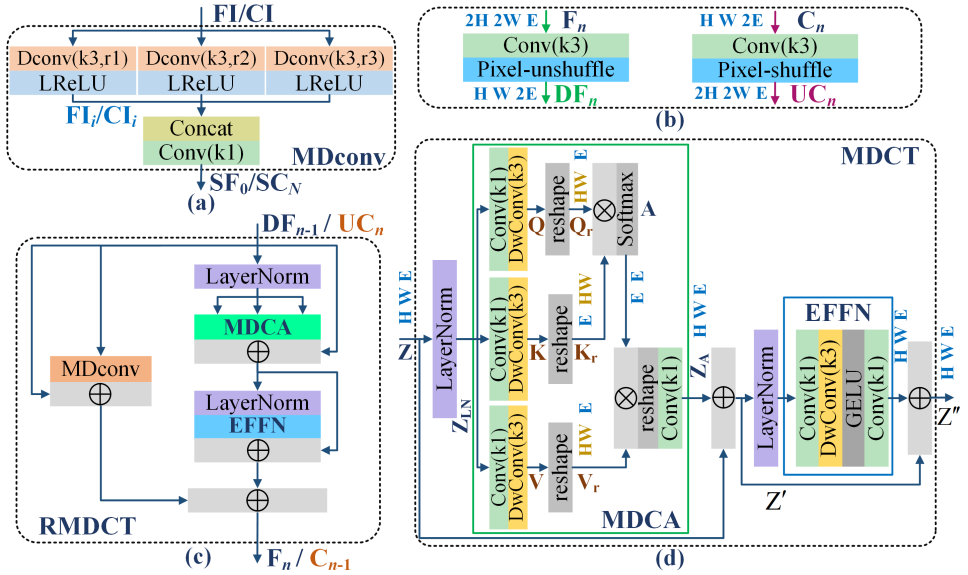


Fig. 3. (a) Dilated convolution (MDonv). (b) Downsampling and upsampling. (c) RMDCT. (d) MDCA and EFFN.

Conv(k3) is applied to adjust the number of channels

$$\mathrm{F}_0 = \kappa_{\mathrm{RMDCT}}(\mathrm{SF}_0) \tag{5}$$

$$\mathrm{F}_n = \kappa_{\mathrm{RMDCT}}(\mathrm{DF}_{n-1}) \quad n = 1, \ldots, N \tag{6}$$

$$\mathrm{DF}_{n-1} = \kappa_{\mathrm{unshuffle}}\left(\kappa^3_{\mathrm{conv}}(\mathrm{F}_{n-1})\right) \quad n = 1, \ldots, N \tag{7}$$

$$\mathrm{C}_N = \kappa_{\mathrm{RMDCT}}(\mathrm{SC}_N) \tag{8}$$

$$\mathrm{C}_{n-1} = \kappa_{\mathrm{RMDCT}}(\mathrm{UC}_n) \quad n = 1, \ldots, N \tag{9}$$

$$\mathrm{UC}_n = \kappa_{\mathrm{shuffle}}\left(\kappa^3_{\mathrm{conv}}(\mathrm{C}_n)\right) \quad n = 1, \ldots, N \tag{10}$$

where F_n and C_n indicate the nth-resolution features of the FI and CI images derived from the RMDCT, respectively; DF_n is acquired by performing a downsampling operation on F_n; UC_n is attained by conducting an upsampling operation on C_n; and $\kappa_{\mathrm{unshuffle}}(\cdot)$ and $\kappa_{\mathrm{shuffle}}(\cdot)$ stand for downsampling and upsampling, respectively; and $\kappa^3_{\mathrm{conv}}(\cdot)$ refers to the convolution with a kernel of 3.

The architecture of the RMDCT is illustrated in Fig. 3(c), which comprises the residual MDconv and a multihead depthwise convolutional Transformer (MDCT) forming the residual structure. The MDCT consists of layer normalization (LayerNorm), multihead depthwise convolutional attention (MDCA), an enhanced feedforward network (EFFN), and residual connections. The differences with Restormer [58] are residual MDconv, EFFN, and residual architecture. RMDCT utilizes the residual MDconv as a direct connection

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in the residual architecture to capture local multiscale information, which exploits a residual connection to avoid gradient vanishing and enhances the representation of local information. The limitation of the local receptive field of CNNs may lead to neglecting the global structure. Therefore, the MDCT is introduced to collaboratively construct the long-range dependency within the image. MDCA in MDCT acquires local–global information via depthwise convolution (DwC) and self-attention. The key character of MDCA is that 1×1 convolutions are utilized to aggregate pixel-level cross-channel context, and 3×3 DwC encodes the channel-wise spatial context. EFFN also enhances local information via the 1×1 convolution and the 3×3 DwC. The MDCA is illustrated in Fig. 3(d). First, implement the LayerNorm on the image Z to acquire $\mathrm{Z_{LN}}$, as displayed in expression (11). Then, 1×1 convolutions are utilized to aggregate pixel-level cross-channel context and 3×3 DwC encodes the channel-wise spatial context generating query (Q), key (K), and value (V), as shown in expression (12). Next, the relevance between Q and K is calculated to gain the global attention weight, i.e., Q and K $\in \mathbb{R}^{H\times W\times E}$ are reshaped to obtain $\mathrm{Q_r} \in \mathbb{R}^{HW\times E}$ and $\mathrm{K_r} \in \mathbb{R}^{E\times HW}$, and the dot product and Softmax operations are conducted to generate the transposed attention map A$\in \mathbb{R}^{E\times E}$, as shown in expression (13). The reshaped $\mathrm{V_r} \in \mathbb{R}^{HW\times E}$ is multiplied by A$\in \mathbb{R}^{E\times E}$ to yield rectified features $\in \mathbb{R}^{HW\times E}$, contributing to the valuable local context; reshape and employ 1×1 convolution to implement feature interconnection and yield feature $\mathrm{Z_A} \in \mathbb{R}^{H\times W\times E}$, defined as in expression (14)

$$\mathrm{Z_{LN}} = \kappa_{\mathrm{LN}}(\mathrm{Z}) \tag{11}$$

$$[\mathrm{Q, K, V}] = \left[\mathrm{dwc_Q^3}\left(c_{\mathrm{Q}}^1(\mathrm{Z_{LN}})\right), \mathrm{dwc_K^3}\left(c_{\mathrm{K}}^1(\mathrm{Z_{LN}})\right)\right.$$
$$\left.\mathrm{dwc_V^3}\left(c_{\mathrm{V}}^1(\mathrm{Z_{LN}})\right)\right] \tag{12}$$

$$\mathrm{A} = \mathrm{Softmax}(\mathrm{K_r} \cdot \mathrm{Q_r}/\varpi\ ) \tag{13}$$

$$\mathrm{Z_A} = \kappa_{\mathrm{conv}}^1(\kappa_{\mathrm{r}}(\mathrm{V_r} \cdot \mathrm{A})) \tag{14}$$

$$\mathrm{Z'} = \mathrm{Z} + \mathrm{Z_A} \tag{15}$$

where κ_{LN} is the LayerNorm operation; Z and $\mathrm{Z_{LN}}$ are the input and the normalized feature, respectively; $\mathrm{Z_A}$ represents the rectified feature; $\mathrm{Z'}$ is the output via the LayerNorm, MDCA, and residual connection; $\mathrm{dwc_Q^3}$ and c_{Q}^1 denote the 3×3 depth-wise convolution and 1×1 convolution for generating Q, respectively; ϖ is a learnable parameter to adjust the amplitude; and κ_{conv}^1 indicates the 1×1 convolution.

In Fig. 3(d), $\mathrm{Z'}$ is delivered into the EFFN and residual connection, which outputs $\mathrm{Z''}$, expressed as (16). Moreover, the DwC is utilized in the EFFN to strengthen the local context

$$\mathrm{Z''} = \kappa_{\mathrm{conv}}^1\left(\kappa_{\mathrm{GELU}}\left(\kappa_{\mathrm{dwconv}}^3\left(\kappa_{\mathrm{conv}}^1\left(\kappa_{\mathrm{LN}}\left(\mathrm{Z'}\right)\right)\right)\right)\right) + \mathrm{Z'} \tag{16}$$

where κ_{GELU} is the Gaussian error linear unit (GELU) function.

With the collaboration of residual MDconv and MDCT, RMDCT can capture local features, establish long-range dependencies, and apply to large-scale images, which obtains local–global features of the CI image (the MS image for SSF and the MODIS image at the prediction time for STF) and the FI image (the PAN image for SSF and the Landsat image at the prior time for STF) to obtain more effective remote sensing information.

2) Multihead Cross-Attention Fusion Decoder: Multiresolution information of the CI and FI extracted by the BALGE only contains the characteristics of a single sensor, and RSIs captured by different sensors with different imaging mechanisms exhibit distinct resolutions and incomplete coverage of bands. The relationship between coarse- and fine-resolution images in STF and SSF is complicated. Therefore, the multiresolution local–global features derived from the BALGE are fused stepwise from low resolution to high resolution by the MCAFD to yield an ideal resolution image, as illustrated in the middle branch of Fig. 2. MCAFD is composed of MFF and MCAFT. To fully exploit the shallow local features $\mathrm{SF_0}$ and SC_N derived from the FI and CI, MFF is introduced to integrate the respective-scale image of $\mathrm{SF_0}$ and SC_N into each stage for image decoding to strengthen the restoration of features. To reduce the sensor bias and exploit the latent relations between dual-source images, MCAFT is introduced to build cross-attention maps of dual-source features for adaptive local–global pixel interactions, which automatically and selectively reinforces dual-source cross-resolution-hierarchy features and aggregates local–global complementary information to recover effective information.

As depicted in Fig. 2, first, downsampling operations are executed on $\mathrm{SF_0}$ to acquire the multiresolution feature $\mathrm{SF}_n \in \mathbb{R}^{(1/2^n)H\times(1/2^n)W\times E_n}$ ($n = 0, 1, \ldots, N$ and $E_n = 32 \times 2^n$); upsampling operations are executed on SC_N to acquire the multiresolution feature $\mathrm{SC}_n \in \mathbb{R}^{(1/2^n)H\times(1/2^n)W\times E_n}$ ($n = 0, 1, \ldots, N$). Then, SF_n and SC_n ($n = 0, 1, \ldots, N$) are skip-connected to the corresponding-resolution $\mathrm{MFF_F}$ and $\mathrm{MFF_C}$ modules of the decoder to perform cross-resolution-hierarchy fusion, respectively. Details of the implementation are given in Fig. 4, and $\mathrm{F}_n \in \mathbb{R}^{(1/2^n)H\times(1/2^n)W\times E_n}$ and $\mathrm{SF}_n \in \mathbb{R}^{(1/2^n)H\times(1/2^n)W\times E_n}$ ($n = 0, 1, \ldots, N$) are concatenated along channels and fused using the pixel-wise convolution to yield $\mathrm{F'}_n \in \mathbb{R}^{(1/2^n)H\times(1/2^n)W\times E_n}$, described as (17). Similarly, $\mathrm{C}_n \in \mathbb{R}^{(1/2^n)H\times(1/2^n)W\times E_n}$ and $\mathrm{SC}_n \in \mathbb{R}^{(1/2^n)H\times(1/2^n)W\times E_n}$ ($n = 0, 1, \ldots, N$) are fused to yield $\mathrm{C'}_n \in \mathbb{R}^{(1/2^n)H\times(1/2^n)W\times E_n}$, described as (18)

$$\mathrm{F'}_n = \kappa_{\mathrm{convl}}^1(\kappa_{\mathrm{concat}}(\mathrm{F}_n, \mathrm{SF}_n)) \quad n = 0, 1, \ldots, N \tag{17}$$

$$\mathrm{C'}_n = \kappa_{\mathrm{convl}}^1(\kappa_{\mathrm{concat}}(\mathrm{C}_n, \mathrm{SC}_n)) \quad n = 0, 1, \ldots, N \tag{18}$$

where $\mathrm{F'}_n$ and $\mathrm{C'}_n$ are the initial fusion results of the $\mathrm{MFF_F}$ and $\mathrm{MFF_C}$ blocks, respectively.

The nth-resolution initial fusion features $\mathrm{F'}_n$, $\mathrm{C'}_n$, and the $(n+1)$th-resolution fusion result $\mathrm{Y}_{n+1} \in \mathbb{R}^{(1/2^{n+1})H\times(1/2^{n+1})W\times E_{n+1}}$ are transferred to the MCAFT performing cross-attention fusion to produce the nth-resolution fusion result $\mathrm{Y}_n \in \mathbb{R}^{(1/2^n)H\times(1/2^n)W\times E_n}$. By duplicating $N+1$ times the aforementioned procedure in the MCAFD, the desired-resolution fusion feature $\mathrm{Y}_0 \in \mathbb{R}^{H\times W\times 32}$ is generated, which then undergoes the RU to produce the final high-resolution image $\mathrm{HI}_0 \in \mathbb{R}^{H\times W\times B}$. The implementation procedures are given as follows:

$$\mathrm{Y'}_n = \kappa_{\mathrm{shuffle}}\left(\kappa_{\mathrm{conv}}^3(\mathrm{Y}_n)\right) \quad n = 1, \ldots, N \tag{19}$$

$$\mathrm{Y}_n = \kappa_{\mathrm{MCAFT}}\left(\mathrm{F'}_n, \mathrm{Y'}_{n+1}, \mathrm{C'}_n\right) \quad n = 0, 1, \ldots, N-1 \tag{20}$$

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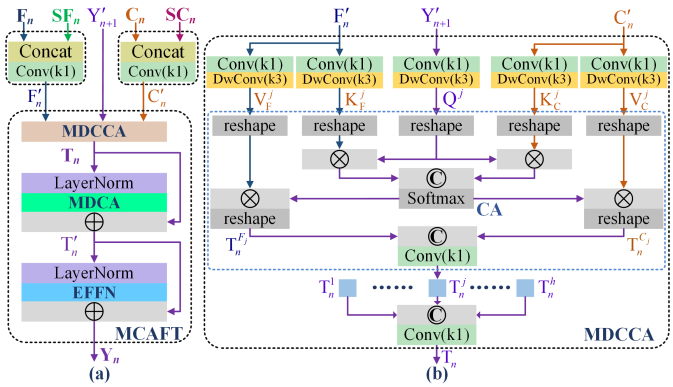


Fig. 4. (a) MFF and MCAFT. (b) Multihead depthwise convolutional cross-attention (MDCCA).

$$\mathrm{Y}_N = \kappa_{\mathrm{MCAFT}}\left(\mathrm{F}'_N, \mathrm{C}'_N\right) \tag{21}$$

$$\mathrm{HI}_n = \kappa^3_{\mathrm{conv}}\left(\kappa^3_{\mathrm{convl}}\left(\kappa^3_{\mathrm{convl}}\left(\mathrm{Y}_n\right)\right)\right) \quad n = 0, 1, \ldots, N \tag{22}$$

where $\mathrm{Y}'_{n+1} \in \mathbb{R}^{(1/2^n)H\times(1/2^n)W\times E_n}$ denotes the upsampled image of Y_{n+1}; Y_n and HI_n are the intermediate fusion outcome and the reconstructed outcome, respectively; $\kappa_{\mathrm{MCAFT}}(\cdot)$ represents the function of MCAFT; and $\kappa^3_{\mathrm{convl}}(\cdot)$ expresses the convolution with a kernel of 3 followed by the LReLU function.

The specific implementation details of MCAFT are displayed in Fig. 4(a). MCAFT is composed of MDCCA and MDCT. MDCCA adaptively integrates dual-source features by capturing the interaction dependencies of the dual-source features via cross-attention with DwC. The blended features with dual-source image properties undergo MDCT to gain local–global features with dual-source image properties. Specifically, for the nth-resolution ($n = 0, 1, \ldots, N-1$), F'_n, C'_n, and $\mathrm{Y}'_{n+1} \in \mathbb{R}^{(1/2^n)H\times(1/2^n)W\times E_n}$ are transmitted to MDCCA to perform cross-attention fusion and acquire $\mathrm{T}_n \in \mathbb{R}^{(1/2^n)H\times(1/2^n)W\times E_n}$. The jth head of MDCCA is illustrated in Fig. 4(b). F'_n undergoes pixel-wise convolutions and DwCs to yield the jth-head $\mathrm{V}^j_{\mathrm{F}}$ and $\mathrm{K}^j_{\mathrm{F}} \in \mathbb{R}^{(1/2^n)H\times(1/2^n)W\times(E_n/h)}$. Meanwhile, the C'_n undergoes pixel-wise convolutions and DwCs to yield the jth-head $\mathrm{V}^j_{\mathrm{C}}$ and $\mathrm{K}^j_{\mathrm{C}} \in \mathbb{R}^{(1/2^n)H\times(1/2^n)W\times(E_n/h)}$. Moreover, applying pixel-wise convolutions and DwCs on Y'_{n+1} produces the jth-head Q^j. The above is accomplished as described in (23)–(25). Q^j and $\mathrm{K}^j_{\mathrm{F}}$ are reshaped and multiplied yielding a matrix $\in ((E_n/h)\times(E_n/h))$. Similarly, Q^j and $\mathrm{K}^j_{\mathrm{C}}$ undergo the above operations obtaining a matrix $\in ((E_n/h)\times(E_n/h))$; the two matrices undergo concatenation and Softmax operations gaining a cross-attention map $\mathrm{CA} \in ((E_n/h)\times 2(E_n/h))$. The reshaped $\mathrm{V}^j_{\mathrm{F}}$ and $\mathrm{V}^j_{\mathrm{C}}$ are multiplied with CA to gain the reinforced images $\mathrm{T}^{\mathrm{F}j}_n$ and $\mathrm{T}^{\mathrm{C}j}_n \in \mathbb{R}^{(1/2^n)H\times(1/2^n)W\times 2(E_n/h)}$, respectively, as described in (26) and (27), as shown at the bottom of the next page. Performing concatenation and fusion operations on $\mathrm{T}^{\mathrm{F}j}_n$ and $\mathrm{T}^{\mathrm{C}j}_n$ obtains the jth-head cross-fusion attention $\mathrm{T}^j_n \in \mathbb{R}^{(1/2^n)H\times(1/2^n)W\times(E_n/h)}$, as presented in (28), as shown at the bottom of the next page. Finally, the h-head cross-fusion attention features are concatenated and projected to obtain the cross-fusion attention feature $\mathrm{T}_n \in \mathbb{R}^{(1/2^n)H\times(1/2^n)W\times E_n}$, as illustrated in (29), as shown at the bottom of the next page

$$\mathrm{Q}^j = \mathrm{dwc}^3_{\mathrm{Q}_j}\left(c^1_{\mathrm{Q}_j}\left(\mathrm{Y}'_{n+1}\right)\right) \tag{23}$$

$$\left[\mathrm{K}^j_{\mathrm{F}}, \mathrm{V}^j_{\mathrm{F}}\right] = \left[\mathrm{dwc}^3_{\mathrm{K}^j_{\mathrm{F}}}\left(c^1_{\mathrm{K}^j_{\mathrm{F}}}\left(\mathrm{F}'_n\right)\right), \mathrm{dwc}^3_{\mathrm{V}^j_{\mathrm{F}}}\left(c^1_{\mathrm{V}^j_{\mathrm{F}}}\left(\mathrm{F}'_n\right)\right)\right] \tag{24}$$

$$\left[\mathrm{K}^j_{\mathrm{C}}, \mathrm{V}^j_{\mathrm{C}}\right] = \left[\mathrm{dwc}^3_{\mathrm{K}^j_{\mathrm{C}}}\left(c^1_{\mathrm{K}^j_{\mathrm{C}}}\left(\mathrm{C}'_n\right)\right), \mathrm{dwc}^3_{\mathrm{V}^j_{\mathrm{C}}}\left(c^1_{\mathrm{V}^j_{\mathrm{C}}}\left(\mathrm{C}'_n\right)\right)\right] \tag{25}$$

where $\mathrm{dwc}^3_{\mathrm{Q}_j}$ and $c^1_{\mathrm{Q}_j}$ are the 3×3 DwC and 1×1 convolution for mapping to Q^j, respectively. When $n = N$, $\mathrm{Y}'_{n+1} = \mathrm{F}'_N$ or $\mathrm{Y}'_{n+1} = \mathrm{C}'_N$ in (23).

Next, T_n is transferred to the MDCT to further decode features and determine complementary information for recovery of valuable information, as described in the following equation:

$$\begin{cases} \mathrm{T}'_n = \kappa_{\mathrm{MDCA}}\left(\kappa_{\mathrm{LN}}\left(\mathrm{T}_n\right)\right) + \mathrm{T}_n \\ \mathrm{Y}_n = \kappa_{\mathrm{EFFN}}\left(\kappa_{\mathrm{LN}}\left(\mathrm{T}'_n\right)\right) + \mathrm{T}'_n \end{cases} \tag{30}$$

where κ_{MDCA} and κ_{EFFN} indicate the MDCA and EFFN operations, respectively; T'_n is obtained by the LayerNorm, MDCA, and residual connection; and Y_n is the nth-resolution intermediate fused image.

As illustrated in Fig. 2, the ideal resolution fusion feature $\mathrm{Y}_0 \in \mathbb{R}^{H\times W\times 32}$ is produced after $N+1$ times of the above fusion and the intermediate fusion feature $\mathrm{Y}_n \in \mathbb{R}^{(1/2^n)H\times(1/2^n)W\times E_n}$ ($n = 1, 2, \ldots, N$) is obtained. Y_n is transferred to the RU to yield the nth-resolution image $\mathrm{HI}_n \in \mathbb{R}^{(1/2^n)H\times(1/2^n)W\times B}$, expressed as in (22), and $\mathrm{HI}_0 \in \mathbb{R}^{H\times W\times B}$ is the final predicted image.

C. Multiresolution Convolutional Transformer Discriminators

Since objects in RSIs vary in size and local–global information is also introduced in the ACTG, motivated by the PatchGAN [60], MCTDs are devised to adversely learn various resolution information for enhancing the realism of the generated image. As depicted in Fig. 5, $N+1$ ($N = 4$) discriminators are devised according to the resolution in the ACTG to learn various resolution information. The inputs of CTD_0–CTD_N are $\mathrm{R}_0/\mathrm{HI}_0$, …, $\mathrm{R}_N/\mathrm{HI}_N$, where R_n ($n = 0, \ldots, N$) signifies the reduced-resolution image obtained by downsampling the reference image R_0 with a factor 2^n; HI_n ($n = 0, \ldots, N$) denotes the nth-resolution image predicted by the ACTG.

First, the CTD_0 extracts the primary local–global feature using the 3×3 convolution and MDCT. Then, depth local–global features are obtained via three convolutions (kernel: 4 and stride: 2) and MDCTs, and the size of features successively decreases. Finally, a result with a dimension of $30\times 30\times 1$ is achieved via two convolutions (kernel: 4, stride: 1) and a sigmoid function. CTD_1 and CTD_2 are identical to the construction of CTD_0, but weights are not shared, yielding results with dimensions of $14\times 14\times 1$ and $6\times 6\times 1$, respectively. Because the input size is smaller, CTD_3 and CTD_4 decrease a convolution (kernel: 4 and stride: 2), and an MDCT and other components are identical to the architecture of CTD_0. CTD_3 and CTD_4 produce outputs with dimensions of $6\times 6\times 1$ and $2\times 2\times 1$, respectively. Meanwhile, a spectral

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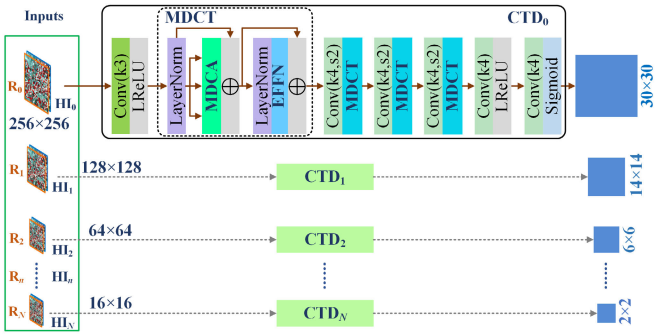


Fig. 5. Architecture of MCTDs.

normalization (SN) operation is introduced following each convolution to accelerate the model convergence and enhance the stability and performance [61]. Moreover, the relative average least-squares (RaLS) discriminator [62], [63] is utilized to judge the relative reality of images to improve model robustness and performance.

D. Generalized Loss Function

Besides the RaLS loss of the CTUGAN, a spatial consistency (SaC) loss, a spectral consistency (SeC) loss, and feature losses, which contain a feature-matching (FM) loss [64] on the MCTDs and a perceptual loss (PL) on the VGG-19 model [65], are introduced to make the CTUGAN fast convergence, steady training, and optimal performance.

The multiresolution adversarial loss of the CTUGAN is presented in the following equation:

$$\min_{G} \max_{D_0, D_1, D_2, D_3, D_4} \sum_{n=0,1,2,3,4} \mathcal{L}_{\text{CTUGAN}}(G, D_n). \tag{31}$$

Expressions for the RaLS function of the ACTG and MCTDs are listed in the following equations:

$$\mathcal{L}_{\text{ACTG}}^{\text{RaLS}} = \sum_{n=0}^{4} \Big\{ \mathbb{E}_{R_n \sim \mathcal{R}} \Big[\big(C_n(R_n) - \mathbb{E}_{P_n \sim \mathcal{P}}[C_n(P_n)] + 1 \big)^2 \Big] + \mathbb{E}_{P_n \sim \mathcal{P}} \Big[\big(C_n(P_n) - \mathbb{E}_{R_n \sim \mathcal{R}}[C_n(R_n)] - 1 \big)^2 \Big] \Big\} \tag{32}$$

$$\mathcal{L}_{\text{MCTD}}^{\text{RaLS}} = \sum_{n=0}^{4} \Big\{ \mathbb{E}_{R_n \sim \mathcal{R}} \Big[\big(C_n(R_n) - \mathbb{E}_{P_n \sim \mathcal{P}}[C_n(P_n)] - 1 \big)^2 \Big] + \mathbb{E}_{P_n \sim \mathcal{P}} \Big[\big(C_n(P_n) - \mathbb{E}_{R_n \sim \mathcal{R}}[C_n(R_n)] + 1 \big)^2 \Big] \Big\}. \tag{33}$$

Moreover,

$$D_n^{\text{RaLS}}(R_n, P_n) = \sigma\big(C_n(R_n) - \mathbb{E}_{P_n \sim \mathcal{P}}[C_n(P_n)] \big) \tag{34}$$

$$D_n^{\text{RaLS}}(P_n, R_n) = \sigma\big(C_n(P_n) - \mathbb{E}_{R_n \sim \mathcal{R}}[C_n(R_n)] \big) \tag{35}$$

where $\mathcal{L}_{\text{ACTG}}^{\text{RaLS}}$ signifies the adversarial loss of the ACTG; $\mathcal{L}_{\text{MCTD}}^{\text{RaLS}}$ indicates the adversarial loss of MCTDs; $D_n^{\text{RaLS}}(\cdot)$ expresses the relative probability of CTD$_n$; σ stands for the sigmoid function; $C_n(\cdot)$ means the untransformed output of CTD$_n$; $\mathcal{R}$ represents a distribution of the nth-resolution reference image R_n and the R_0 image has the maximum resolution; $\mathcal{P}$ represents a distribution of the nth-resolution predicted image P_n; $\mathbb{E}_{R_n \sim \mathcal{R}}$ and $\mathbb{E}_{P_n \sim \mathcal{P}}$ depict the mean operation on R_n and P_n images in a batch, respectively.

The FM loss is a learned similarity indicator quantified by the difference in features on MCTDs between the predicted and reference images, which helps to optimize the ACTG and improve model stability. The FM loss utilizes the L1 [66] distance of R_n and P_n features on the MCTDs, formulated as follows:

$$\mathcal{L}_{\text{FM}}(G, D_n) = \mathbb{E}_{(R_n \sim \mathcal{R}, P_n \sim \mathcal{P})} \left[\sum_{l=1}^{\mathbb{Z}} \frac{1}{M_l} \left(\left\| D_n^l(R_n) - D_n^l(P_n) \right\|_1 \right) \right] \tag{36}$$

where $\mathcal{L}_{\text{FM}}(G, D_n)$ is the FM loss on the CTD$_n$, $\mathbb{Z}$ is the number of layers, D_n^l is the lth-layer feature on the CTD$_n$, and M_l is the number of features of the lth layer.

The PL is the feature difference between the generated image and the reference image expressed on the pretrained model; the smaller the loss, the closer the two are. Similar to ESRGAN [67], minimize the distance between the two features before the activation function. $\mathcal{L}_{\text{PL}}$ is reported in the following equation:

$$\mathcal{L}_{\text{PL}} = \frac{1}{S} \sum_{\alpha=1}^{S} (f(R_0) - f(P_0))^2 \tag{37}$$

where $\mathcal{L}_{\text{PL}}$ denotes the PL function; S means the batch size; and $f(R_0)$ and $f(P_0)$ represent features of R_0 and P_0 on the pretrained VGG-19 model [65], respectively.

High-frequency information and gradient information in an image express spatial information such as edges and details. To maintain the consistency of the spatial information between the predicted and reference images, the high-frequency loss and the gradient loss constitute the SaC loss $\mathcal{L}_{\text{SaC}}$, given in

$$\mathrm{T}_n^{\mathrm{F}_j} = \kappa_r \left(\kappa_r \left(\mathrm{V}_{\mathrm{F}}^j \right) \cdot \text{Softmax} \left(\frac{\kappa_{\text{concat}} \left(\kappa_r \left(\mathrm{K}_{\mathrm{F}}^j \right) \kappa_r \left(\mathrm{Q}^j \right), \kappa_r \left(\mathrm{K}_{\mathrm{C}}^j \right) \kappa_r \left(\mathrm{Q}^j \right) \right)}{\varpi} \right) \right) \tag{26}$$

$$\mathrm{T}_n^{\mathrm{C}_j} = \kappa_r \left(\kappa_r \left(\mathrm{V}_{\mathrm{C}}^j \right) \cdot \text{Softmax} \left(\frac{\kappa_{\text{concat}} \left(\kappa_r \left(\mathrm{K}_{\mathrm{F}}^j \right) \kappa_r \left(\mathrm{Q}^j \right), \kappa_r \left(\mathrm{K}_{\mathrm{C}}^j \right) \kappa_r \left(\mathrm{Q}^j \right) \right)}{\varpi} \right) \right) \tag{27}$$

$$\mathrm{T}_n^j = \kappa_{\text{conv}}^1 \left(\kappa_{\text{concat}} \left(\mathrm{T}_n^{\mathrm{F}_j}, \mathrm{T}_n^{\mathrm{C}_j} \right) \right) \tag{28}$$

$$\mathrm{T}_n = \kappa_{\text{conv}}^1 \left(\kappa_{\text{concat}} \left(\mathrm{T}_n^1, \ldots, \mathrm{T}_n^j, \ldots, \mathrm{T}_n^h \right) \right) \tag{29}$$

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the following equation:

$$\mathcal{L}_{\mathrm{SaC}} = \frac{1}{S}\sum_{\alpha=1}^{S}\|h(R_0) - h(P_0)\|_F^2 + \frac{1}{S}\sum_{\alpha=1}^{S}\|\nabla(R_0) - \nabla(P_0)\|_F^2 \quad (38)$$

where $h(\cdot)$ and $\nabla(\cdot)$ extract the high-frequency information and gradient information, respectively; and F means the F-norm.

The source image contains abundant spectral information. To maintain the consistency of spectra between the predicted and reference images, the loss between the predicted and source images denotes the SeC loss $\mathcal{L}_{\mathrm{SeC}}$, given in the following equation:

$$\mathcal{L}_{\mathrm{SeC}} = \frac{1}{S}\sum_{\alpha=1}^{S}\left\|\overrightarrow{R} - \overrightarrow{P}\right\|_F^2 \quad (39)$$

where $\overrightarrow{R}$ and $\overrightarrow{P}$ are the reference and generated images used for distinguishing spectra, respectively.

In STF, reference image R_0 is the Landsat image at the predicted date; $\overrightarrow{R} = R_0$ and $\overrightarrow{P} = \mathrm{HI}_0$. In SSF, the reference image R_0 is the captured PAN image; $\overrightarrow{R} = \overrightarrow{R}_2$ and $\overrightarrow{R}_2$ is derived from the reduced-spectral resolution of the captured MS image, expressed as (40); $\overrightarrow{P} = \overrightarrow{\mathrm{HI}}_2$ and $\overrightarrow{\mathrm{HI}}_2$ is obtained by the reduced-spectral resolution of HI_2 image

$$\overrightarrow{\mathrm{X}}_n = \sum_{b=1}^{B}\chi_n^b \cdot \widetilde{\mathrm{MS}}_n^b + \eta_n^b \quad (40)$$

where $\overrightarrow{\mathrm{X}}_n$ stands for the nth-resolution PAN image derived from the reduced-spectral resolution of $\widetilde{\mathrm{MS}}$; B is the number of bands of the $\widetilde{\mathrm{MS}}$ image; and χ_n^b and η_n^b indicate the nth-scale weight and bias, respectively.

Therefore, the total loss function $\mathcal{L}_{\mathrm{A}}$ is described in (41), where μ_1, ρ_2, ς_3, λ_4, and ξ_5 are coefficients

$$\mathcal{L}_{\mathrm{A}} = \mu_1\mathcal{L}_{\mathrm{ACTG}}^{\mathrm{RaLS}} + \rho_2\sum_{n=0}^{4}\mathcal{L}_{\mathrm{FM}}(G, D_n) + \varsigma_3\mathcal{L}_{\mathrm{PL}} + \lambda_4\mathcal{L}_{\mathrm{SaC}} + \xi_5\mathcal{L}_{\mathrm{SeC}}. \quad (41)$$

III. Experiments

A. Datasets

1) SSF Data: The SSF data were collected by GF-2[1] and QuickBird (QB)[2] satellites. Both GF-2 and QB images are four bands and the radiation resolutions are 10 and 11 bit, respectively. The resolutions of MS and PAN images for GF-2 are 4 and 1 m, respectively; the resolutions of MS and PAN images for QB are 2.4 and 0.6 m, respectively. The ratio of spatial resolution of MS and PAN images is both 4.

To fairly compare the supervised and unsupervised SSF models, a reduced-resolution dataset for supervised training and a full-resolution dataset for unsupervised training are generated from the source MS and PAN images following Wald's [68] rule. The supervised SSF models are trained on the reduced-resolution dataset and the source MS image serves as the ground truth. Furthermore, the quantity and dimension of the reduced-resolution data and full-resolution data utilized for training remain the same.

The GF-2 dataset consists of seven raw images, of which one image is assigned for testing and the rest six images are assigned for training. The six images are randomly cropped to create 12 000 samples; 9600 samples are randomly chosen for training and the rest 2400 samples are for validation. Moreover, 286 full-resolution test data are produced.

The QB dataset consists of eight raw images, of which one image is assigned for testing, and the rest seven images are assigned for training. The seven images are randomly cropped to create 8000 samples; 6400 samples are randomly chosen for training and the rest 1600 samples are for validation. Moreover, 117 full-resolution test data are produced.

The dimensions of MS and PAN images utilized for training are $64 \times 64 \times 4$ and $256 \times 256 \times 1$, respectively, and the dimensions of MS and PAN images utilized for testing are $100 \times 100 \times 4$ and $400 \times 400 \times 1$, respectively.

2) STF Data: STF experiments are performed utilizing the Coleambally Irrigation Area (CIA)[3] and lower Gwydir catchment (LGC)[4] data [69]. The CIA data comprise 17 pairs of Landsat–MODIS images. Landsat images were gained by Landsat-7 ETM+. MODIS images were MODIS Terra MOD09GA Collection 5 products. The height and width band of images is $2040 \times 1720 \times 6$. Scenarios in the CIA are of spatial heterogeneity, containing numerous small patches of farmlands with spatial scatter and irregular shapes. CIA scenarios have significant phenological changes.

The LGC data comprise 14 pairs of Landsat–MODIS images. Landsat images were gained by Landsat-5 TM. MODIS images were MODIS Terra MOD09GA Collection 5 products. The dimension of images is $2720 \times 3200 \times 6$. LGC scenarios have significant land-cover-type changes.

The ratio of Landsat–MODIS images is about 16. The dimension of the CIA image for training is $2032 \times 1358 \times 6$ and that of the LGC image is $2708 \times 3184 \times 6$.

B. Experimental Setup

The framework for realizing DL-based models is PyTorch; STARFM and FSDAF approaches are implemented in IDL; and GSA, BDSD, SFIM, and MTF-GLP-HPM approaches are performed under MATLAB. The platform comprises an NVIDIA Tesla A100 PCIe GPU with 40 GiB of video memory and Intel[5] Xeon[5] Gold 6326 CPU @ 2.90 GHz. The dimension of fine-resolution images for training is $256 \times 256 \times C$, the batch size is 8, and the epoch is 200. The setting of hyperparameters in $\mathcal{L}_{\mathrm{A}}$ is $\mu_1 = 1 \times 10^{-3}$, $\rho_2 = \varsigma_3 = 1 \times 10^{-3}$, and $\lambda_4 = \xi_5 = 5 \times 10^{-4}$. An Adam [70] optimizer is applied to optimize the model with an initial learning rate

[1]https://data.cresda.cn/#/home; https://www.cpeos.org.cn/#/?source=public
[2]https://www.maxar.com/
[3]https://data.csiro.au/collection/csiro:5846v3
[4]https://data.csiro.au/collection/csiro:5847v003
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TABLE I
OBJECTIVE ASSESSMENT OF SSF RESULTS ON THE QB DATA

Model	D_λ	D_S	QNR
GSA	0.0495	0.1315	0.8255
BDSD	0.0223	0.0883	0.8913
SFIM	0.0645	0.0730	0.8672
MTF-GLP-HPM	0.0634	0.0720	0.8692
PsGAN	0.0299	0.0150	0.9555
RED-cGAN	0.0194	0.0157	0.9652
DIGAN	0.0209	0.0138	0.9656
PGMAN	0.0123	0.0144	0.9735
UCGAN	0.0109	0.0119	0.9774
PanFormer	0.0326	0.0353	0.9333
BPF-MGAN	0.0061	0.0070	0.9869
CTUGAN	**0.0058**	**0.0063**	**0.9880**
Ideal value	0	0	1

TABLE II
OBJECTIVE ASSESSMENT OF COMPARATIVE RESULTS ON THE GF-2 DATA

Model	D_λ	D_S	QNR
GSA	0.1023	0.1422	0.7700
BDSD	0.0234	0.0724	0.9058
SFIM	0.0843	0.0263	0.8916
MTF-GLP-HPM	0.0985	0.0660	0.8420
PsGAN	0.0083	0.0436	0.9485
RED-cGAN	0.0126	0.0486	0.9394
DIGAN	0.0119	0.0342	0.9543
PGMAN	0.0122	0.0070	0.9809
UCGAN	0.0080	0.0099	0.9822
PanFormer	0.0062	0.0296	0.9644
BPF-MGAN	0.0062	0.0059	0.9880
CTUGAN	**0.0046**	**0.0036**	**0.9918**
Ideal value	0	0	1

of 2×10^{-4}. Experiments based on DL are implemented on the GPU and traditional models are executed on the CPU.

C. SSF Experimental Results

Extensive SSF experiments are carried out on the captured data to validate the CTUGAN performance. SSF is realized on images derived from GF-2 and QB. The proposed CTUGAN model is compared with prevailing SSF methods, where traditional approaches contain the GSA [10], BDSD [12], SFIM [13], and MTF-GLP-HPM [14], and DL models comprise the RED-cGAN [26], PsGAN [27], DIGAN [28], PGMAN [33], UCGAN [34], PanFormer [30], and BPF-MGAN [52]. The experimental outcomes are subjectively and objectively assessed. The common indicators of quality without reference (QNR) comprise a spectral distortion index D_λ, a spatial distortion metric D_S, and QNR [71]. The desired value of QNR is 1 and the desired value of D_λ and D_S is 0.

1) Experiment on the QB Data: The results of comparative experiments conducted on the QB test data are displayed in Fig. 6. The NIR–red–green band serves as the R–G–B channel. Fig. 6(a) shows the raw PAN image with a spatial resolution of 0.6 m; Fig. 6(b) presents the raw MS image with a spatial resolution of 2.4 m; and Fig. 6(c)–(n) illustrates the SSF results of the GSA, BDSD, SFIM, MTF-GLP-HPM, PsGAN, RED-cGAN, DIGAN, PGMAN, UCGAN, PanFormer, BPF-MGAN, and CTUGAN. To further observe details among the comparative approaches, contents in the yellow box of Fig. 6 are magnified. Table I depicts the objective indicators to assess SSF outcomes. Bold values mean optimal and blue values indicate suboptimal. From Fig. 6, it is evident that PsGAN, RED-cGAN, DIGAN, PGMAN, and PanFormer suffer from significant color distortion. The DIGAN, PGMAN, and PanFormer exhibit severe ringing effects. Combined with Table I, it is also noticed that the SFIM and MTF-GLP-HPM spectra are severely distorted, followed by the GSA. From Fig. 6, it is further observed that the UCGAN, BPF-MGAN, and CTUGAN results preserve the spectrum information much closer to the MS image. From the zoomed contents, it is revealed that the textures, details, and spectra of the CTUGAN result are more consistent with the MS and PAN images. Furthermore, in Table I, D_λ, D_S, and QNR of the proposed CTUGAN model are all optimal.

2) Experiment on the GF-2 Data: Fig. 7 exhibits the results of comparative experiments conducted on the GF-2 test data. Fig. 7(a) presents a spatial resolution of 1 m and Fig. 7(b) presents a spatial resolution of 4 m. Contents in the yellow box of Fig. 7 are magnified. Table II depicts the objective indicators to assess SSF outcomes. Severe artifacts in the BDSD, SFIM, and DIGAN results are obviously viewed in Fig. 7. The GSA presents more spectral distortions. The PGMAN result suffers from more severe color distortions. The PsGAN suffers from lacking textures. Together with the enlarged details, it is also noticed that the details and edges of the MTF-GLP-HPM, DIGAN, PGMAN, UCGAN, and BPF-MGAN results are relatively fuzzy; the PsGAN, RED-cGAN, and PanFormer suffer from color distortion on small targets. Overall, the result of the proposed CTUGAN model is optimal for visual assessment. Moreover, in Table II, D_λ, D_S, and QNR of the proposed CTUGAN model are optimal.

The analyses of SSF findings reveal that the proposed CTUGAN model is optimal in both visual and objective evaluations, which verifies that the proposed CTUGAN model enhances the spatial resolution to a greater extent while retaining the spectra.

D. STF Experimental Results

STF experiments are executed on the CIA and LGC data. The proposed CTUGAN model is compared with the prevailing STF models, where traditional models contain the STARFM [40] and FSDAF [42], and DL networks comprise the EDCSTFN [43], GAN-STFM [44], SwinSTFM [46], and BPF-MGAN [52]. The STARFM, FSDAF, EDCSTFN, and SwinSTFM models take three images as inputs, i.e., Landsat–MODIS images at t_0 date and the MODIS image at t_1 date, producing the Landsat image at t_1 date. The GAN-STFM, BPF-MGAN, and CTUGAN models take two images as inputs, i.e., the Landsat image at t_0 date and the MODIS image at t_1 date. The experimental outcomes are subjectively and objectively assessed. The dimension of the CIA data for evaluating is $2032 \times 1344 \times 6$ and that of the LGC image is $2688 \times 3184 \times 6$. Metrics commonly utilized for objective assessment comprise root-mean-square error (RMSE), spectral angle mapper (SAM) [72], structural

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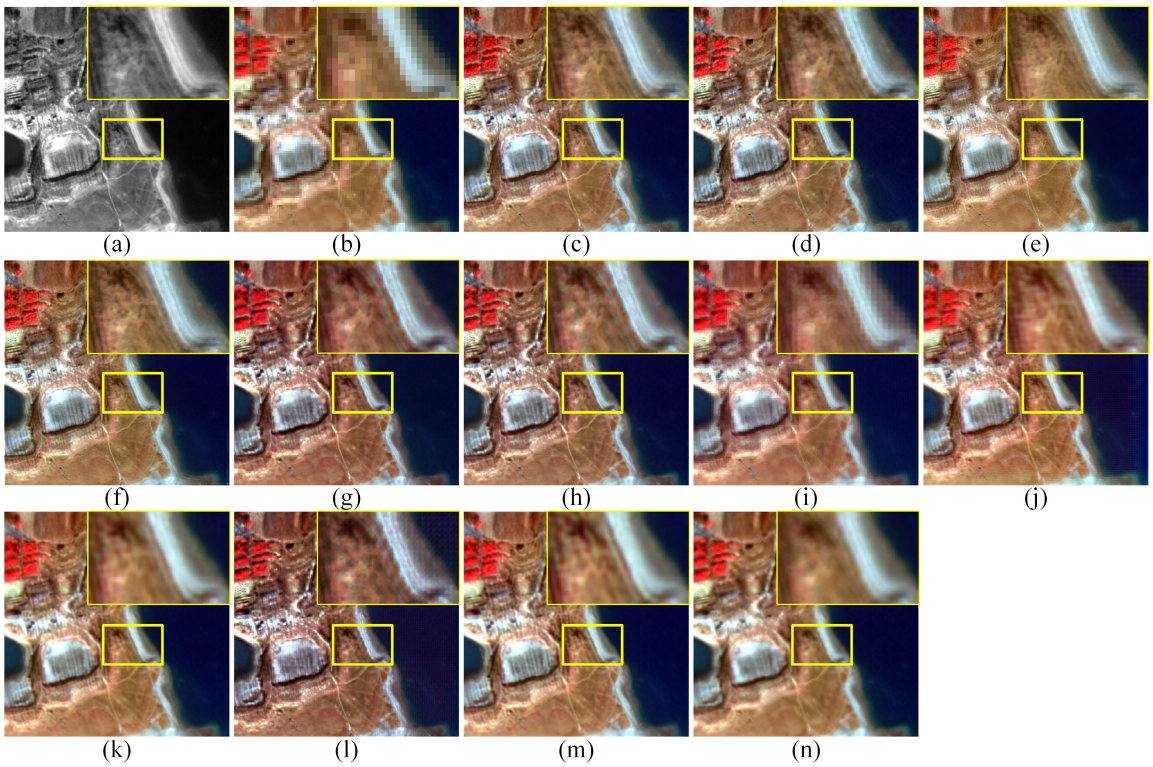


Fig. 6. Visual assessment of comparative results on QB full-resolution data. (a) PAN. (b) MS. (c) GSA. (d) BDSD. (e) SFIM. (f) MTF-GLP-HPM. (g) PsGAN. (h) RED-cGAN. (i) DIGAN. (j) PGMAN. (k) UCGAN. (l) PanFormer. (m) BPF-MGAN. (n) CTUGAN.

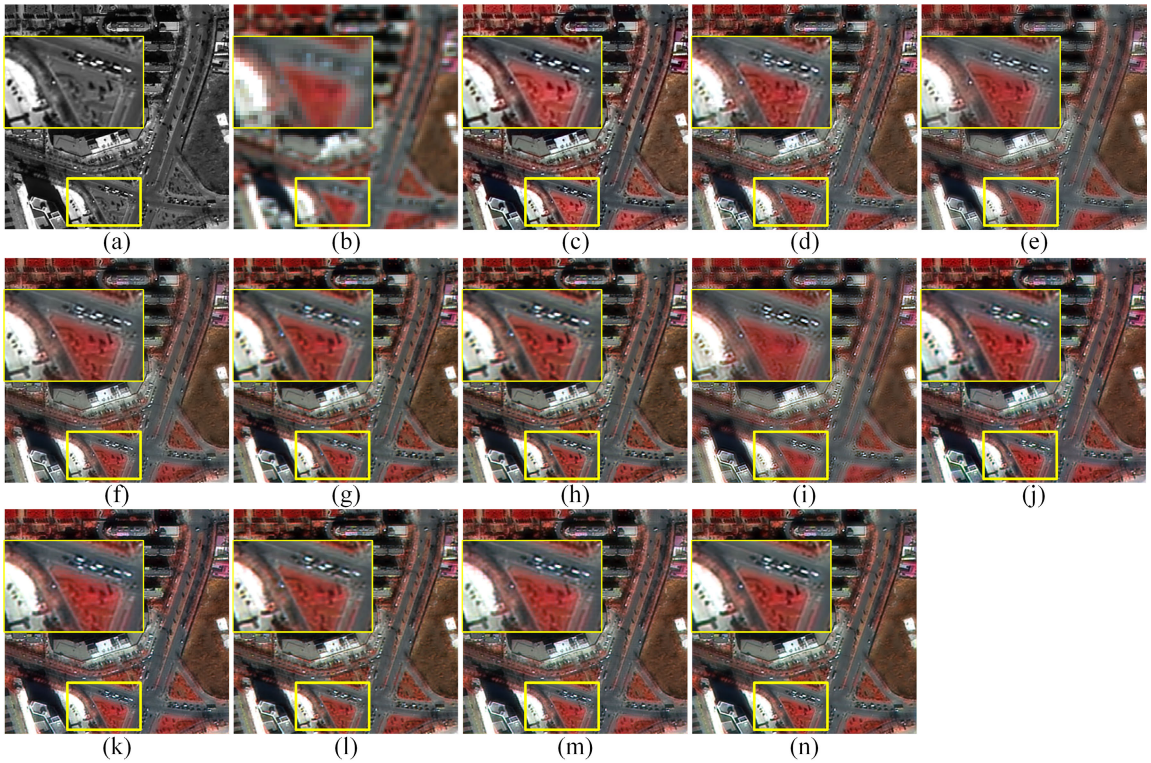


Fig. 7. Visual assessment of comparative results on GF-2 full-resolution data. (a) PAN. (b) MS. (c) GSA. (d) BDSD. (e) SFIM. (f) MTF-GLP-HPM. (g) PsGAN. (h) RED-cGAN. (i) DIGAN. (j) PGMAN. (k) UCGAN. (l) PanFormer. (m) BPF-MGAN. (n) CTUGAN.

similarity (SSIM) [73], erreur relative globale adimensionnelle de synthése (ERGAS) [74], and correlation coefficient (CC) [42]. The desired value for the SSIM and CC is 1; the desired value for the RMSE, SAM, and ERGAS is 0.

1) Experiment on the CIA Data: The objective metrics for assessing the STF results on the CIA test data are listed in Table III. The RMSE, CC, and SSIM are the average values of the individual band, while the ERGAS and SAM

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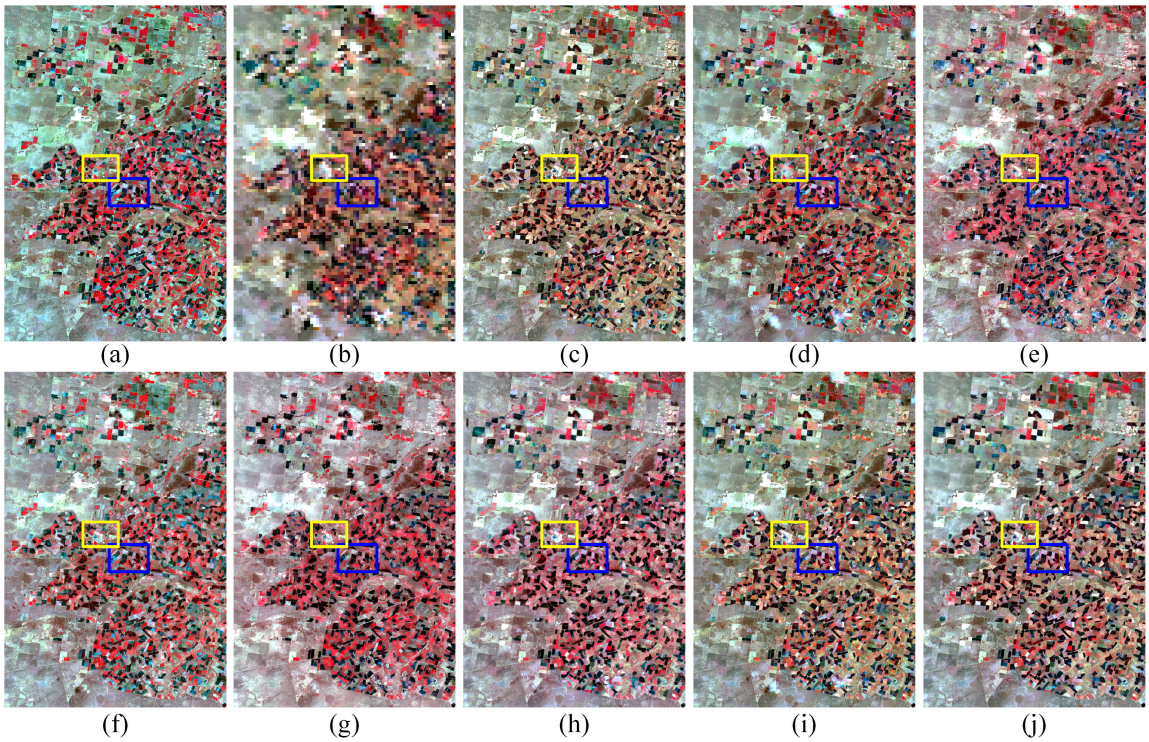


Fig. 8. Predictions for the November 25, 2001, scenario from the CIA. (a) Landsat image on November 9, 2001. (b) MODIS image on November 25, 2001. (c) Landsat image on November 25, 2001 (ground truth). (d) STARFM. (e) FSDAF. (f) EDCSTFN. (g) GAN-STFM. (h) SwinSTFM. (i) BPF-MGAN. (j) CTUGAN.

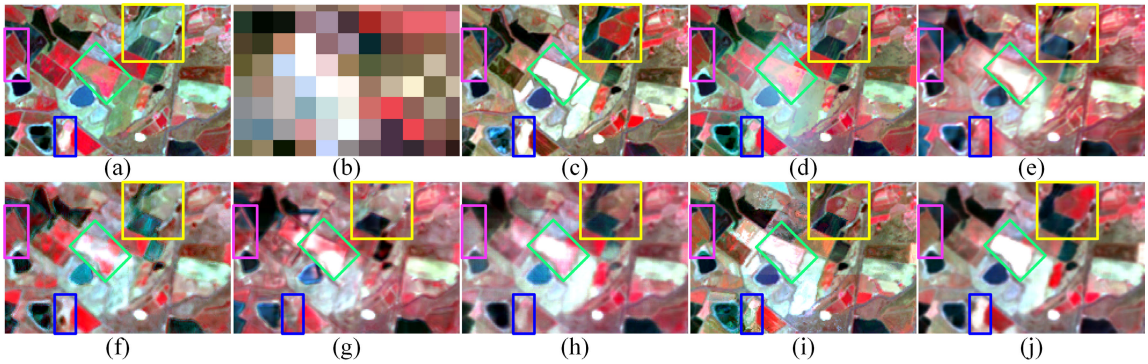


Fig. 9. Magnified contents of the yellow box in Fig. 8. (a) Landsat image on November 9, 2001. (b) MODIS image on November 25, 2001. (c) Landsat image on November 25, 2001 (ground truth). (d) STARFM. (e) FSDAF. (f) EDCSTFN. (g) GAN-STFM. (h) SwinSTFM. (i) BPF-MGAN. (j) CTUGAN.

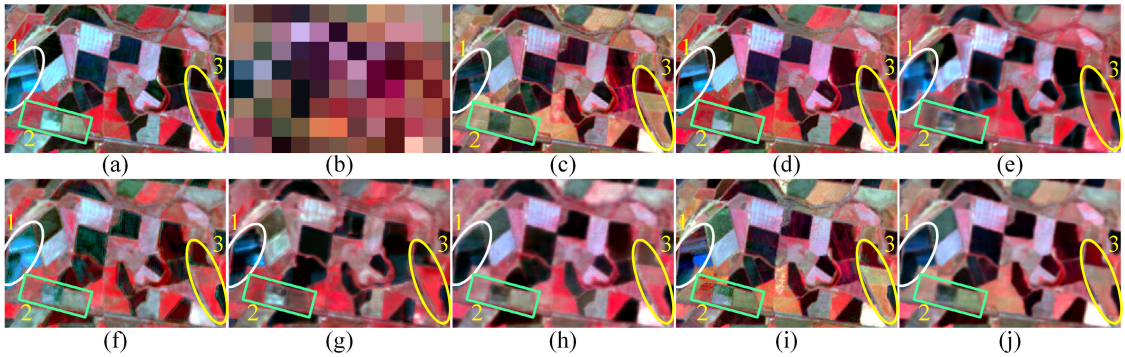


Fig. 10. Magnified contents of the blue box in Fig. 8. (a) Landsat image on November 9, 2001. (b) MODIS image on November 25, 2001. (c) Landsat image on November 25, 2001 (ground truth). (d) STARFM. (e) FSDAF. (f) EDCSTFN. (g) GAN-STFM. (h) SwinSTFM. (i) BPF-MGAN. (j) CTUGAN.

are calculated across all bands. From Table III, it is noted that the predicted outcomes of the CTUGAN model are optimal for radiation, structure, and spectrum fidelity, except for the prediction of scenes on January 5, 2002. For the prediction of the scenario on January 5, 2002, the CTUGAN attains the optimal performance regarding both radiance and spectrum

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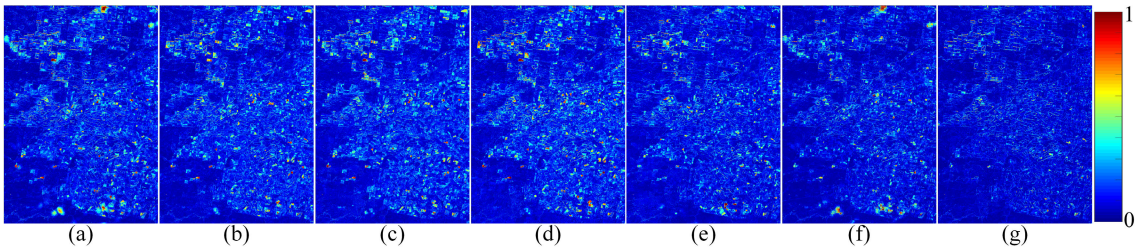


Fig. 11. MAEM between the prediction and ground truth. (a) STARFM. (b) FSDAF. (c) EDCSTFN. (d) GAN-STFM. (e) SwinSTFM. (f) BPF-MGAN. (g) CTUGAN.

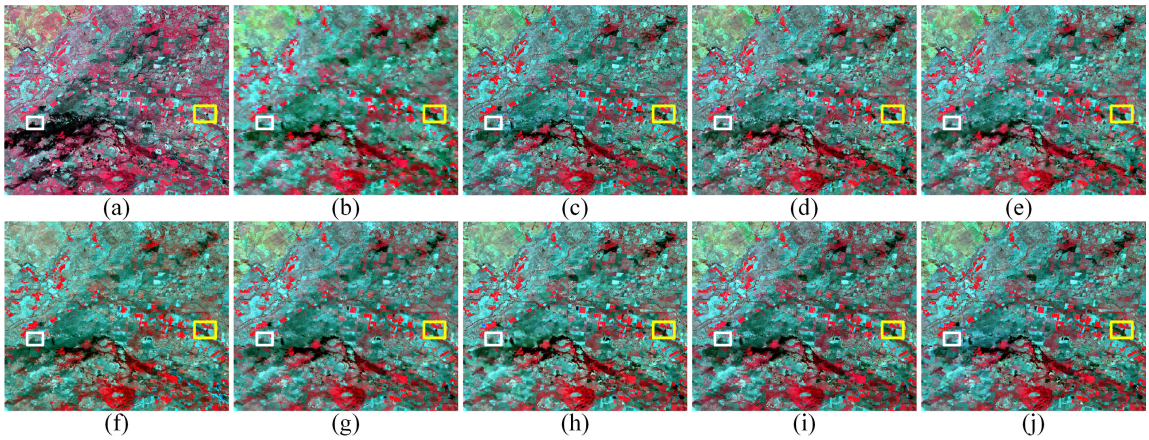


Fig. 12. Predictions of the January 13, 2005, scenario from the LGC. (a) Landsat image on December 28, 2004. (b) MODIS image on January 13, 2005. (c) Landsat image on January 13, 2005, i.e., ground truth. (d) STARFM. (e) FSDAF. (f) EDCSTFN. (g) GAN-STFM. (h) SwinSTFM. (i) BPF-MGAN. (j) CTUGAN.

fidelity, whereas the CC and SSIM rank second behind the BPF-MGAN. It is also revealed that the BPF-MGAN lies in second for all metrics on four scenarios besides the 4th scene, whereas the SwinSTFM only ranks second on the 4th scenario. This also verifies that the CTUGAN is more stable and robust for different scenarios with massive phenological changes.

The CIA scenarios have extensive phenological changes, and the predictions for the November 25, 2001, scenario are presented in Fig. 8. The Landsat and MODIS images on November 9, 2001, are the prior data (the GAN-STFM, BPF-MGAN, and CTUGAN only utilize the Landsat image), and the Landsat image on November 25, 2001, is the ground truth. The NIR–red–green band acts as the R–G–B channel in Fig. 8. Fig. 8(a) depicts the Landsat data acquired on November 9, 2001; Fig. 8(b) shows the MODIS data acquired on November 25, 2001; Fig. 8(c) shows the Landsat data acquired on November 25, 2001; and Fig. 8(d)–(j) illustrates the predictions for Fig. 8(b) produced by the STARFM, FSDAF, EDCSTFN, GAN-STFM, SwinSTFM, BPF-MGAN, and CTUGAN approaches, respectively. To facilitate discovering the differences between comparative methodologies, the yellow and blue boxes in Fig. 8 are enlarged and exhibited in Figs. 9 and 10. The mean absolute error map (MAEM) between the prediction and ground truth is presented in Fig. 11.

Fig. 8 apparently conveys that the STARFM, FSDAF, EDCSTFN, and GAN-STFM results suffer from severe spectral distortions; the predictions of changes favor Fig. 8(a) rather than Fig. 8(c), which shows low precision in forecasting the scene changes. The predictions of scene changes by the BPF-MGAN and CTUGAN approaches are much closer to Fig. 8(c). From Fig. 9, it is more easily seen that the STARFM, FSDAF, EDCSTFN, GAN-STFM, and SwinSTFM are poor at forecasting scenario variations and rely on the prior data. By close inspection, in Fig. 9, the CTUGAN model predicts more precisely than the BPF-MGAN approach for the changes in the pink, blue, green, and yellow boxes. From regions 1 and 2 in Fig. 10, it is noticed that the STARFM, FSDAF, EDCSTFN, and GAN-STFM give the worst predictions for large variations, followed by the BPF-MGAN, and the SwinSTFM and CTUGAN approaches are better, whereas the CTUGAN outcomes are nearer to the ground truth. From region 3 in Fig. 10, it is found that the STARFM, FSDAF, EDCSTFN, GAN-STFM, and SwinSTFM give the worst predictions for large variations, and the BPF-MGAN and CTUGAN approaches are better, whereas the CTUGAN outcome is nearer to the ground truth. Table III demonstrates that all the indexes of the proposed CTUGAN approach are optimal. Fig. 11 presents that the MAEM of the CTUGAN is minimal. In summary, the STF result of the CTUGAN on the CIA data is optimal in both visual assessment and metrics.

2) Experiment on the LGC Data: The objective metrics for assessing the STF results on the LGC test data are listed in Table IV. From Table IV, it is noted that the predicted outcomes of the CTUGAN model are optimal for radiation, structure, and spectrum fidelity. It is also revealed that the metrics of BPF-MGAN are suboptimal for all but the 1st scene, while the metrics of SwinSTFM lie suboptimal only in the 1st scene. This also validates that the proposed CTUGAN

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TABLE III
OBJECTIVE METRICS FOR ASSESSING THE STF RESULTS ON THE CIA TEST DATA

Prediction date	Model	RMSE	ERGAS	SAM	CC	SSIM
20011125	STARFM	0.0295	0.9854	0.1028	0.8140	0.8370
	FSDAF	0.0291	0.9609	0.1070	0.8165	0.8375
	EDCSTFN	0.0293	0.9528	0.1095	0.8227	0.8332
	GAN-STFM	0.0314	1.0297	0.1180	0.8027	0.8212
	SwinSTFM	0.0263	0.8811	0.0911	0.8704	0.8595
	BPF-MGAN	0.0251	0.8667	0.0799	0.8653	0.8715
	CTUGAN	**0.0206**	**0.6640**	**0.0654**	**0.9169**	**0.8830**
20011204	STARFM	0.0294	0.9126	0.0860	0.8641	0.8715
	FSDAF	0.0292	0.8309	0.0889	0.8509	0.8593
	EDCSTFN	0.0305	1.0293	0.0858	0.8621	0.8680
	GAN-STFM	0.0316	0.9273	0.0971	0.8214	0.8417
	SwinSTFM	0.0278	0.8999	0.0797	0.8947	0.8749
	BPF-MGAN	0.0245	0.6838	0.0708	0.9074	0.8918
	CTUGAN	**0.0189**	**0.5264**	**0.0520**	**0.9426**	**0.9073**
20020105	STARFM	0.0381	0.9697	0.1182	0.8080	0.8267
	FSDAF	0.0375	0.9972	0.0859	0.8401	0.8553
	EDCSTFN	0.0369	1.0089	0.0981	0.8306	0.8443
	GAN-STFM	0.0355	0.9275	0.0837	0.8224	0.8136
	SwinSTFM	0.0308	0.8333	0.0698	0.8928	0.8622
	BPF-MGAN	**0.0235**	0.6218	0.0575	**0.9365**	**0.9010**
	CTUGAN	**0.0235**	**0.5962**	**0.0527**	0.9290	0.8912
20020112	STARFM	0.0291	0.7904	0.0824	0.9355	0.8808
	FSDAF	0.0235	0.7232	0.0700	0.9355	0.8956
	EDCSTFN	0.0374	0.9026	0.0755	0.9214	0.8731
	GAN-STFM	0.0279	0.7827	0.0761	0.8936	0.8559
	SwinSTFM	0.0202	0.6000	0.0529	0.9481	0.9129
	BPF-MGAN	0.0215	0.5909	0.0552	0.9428	0.9052
	CTUGAN	**0.0191**	**0.5339**	**0.0458**	**0.9583**	**0.9219**
20020213	STARFM	0.0318	1.0356	0.0987	0.8539	0.8227
	FSDAF	0.0320	1.0085	0.1118	0.8613	0.8391
	EDCSTFN	0.0315	0.9356	0.0918	0.8683	0.8607
	GAN-STFM	0.0255	0.8129	0.0770	0.8990	0.8616
	SwinSTFM	0.0235	0.8306	0.0701	0.9190	0.8807
	BPF-MGAN	0.0220	0.6915	0.0606	0.9283	0.8907
	CTUGAN	**0.0216**	**0.6628**	**0.0587**	**0.9344**	**0.8928**
	Ideal value	0	0	0	1	1

TABLE IV
OBJECTIVE METRICS FOR ASSESSING THE STF RESULTS ON THE LGC TEST DATA

Prediction date	Model	RMSE	ERGAS	SAM	CC	SSIM
20041212	STARFM	0.0586	3.7782	0.3813	0.2116	0.6962
	FSDAF	0.0349	2.2671	0.2069	0.7502	0.7718
	EDCSTFN	0.0505	1.8530	0.2771	0.6442	0.7528
	GAN-STFM	0.0376	1.8018	0.2296	0.7004	0.7453
	SwinSTFM	0.0315	1.6760	0.1815	0.7793	0.8021
	BPF-MGAN	0.0340	1.6613	0.1825	0.7532	0.7840
	CTUGAN	**0.0288**	**1.4591**	**0.1546**	**0.8003**	**0.8233**
20041228	STARFM	0.0337	1.4477	0.1505	0.7763	0.8067
	FSDAF	0.0324	1.3931	0.1395	0.7799	0.7986
	EDCSTFN	0.0409	1.7214	0.1918	0.6364	0.8058
	GAN-STFM	0.0296	1.2346	0.1303	0.7981	0.8276
	SwinSTFM	0.0263	1.1260	0.1153	0.8490	0.8496
	BPF-MGAN	0.0252	0.9847	0.1069	0.8639	0.8628
	CTUGAN	**0.0234**	**0.9661**	**0.1003**	**0.8776**	**0.8682**
20050113	STARFM	0.0232	0.7959	0.0840	0.8739	0.8921
	FSDAF	0.0217	0.7505	0.0809	0.8815	0.8879
	EDCSTFN	0.0223	0.8069	0.0812	0.8619	0.9014
	GAN-STFM	0.0201	0.6581	0.0668	0.9068	0.9160
	SwinSTFM	0.0188	0.6455	0.0692	0.9217	0.9166
	BPF-MGAN	0.0182	0.5953	0.0638	0.9237	0.9295
	CTUGAN	**0.0148**	**0.5158**	**0.0540**	**0.9405**	**0.9299**
20050129	STARFM	0.0252	0.8837	0.0775	0.8973	0.8942
	FSDAF	0.0254	0.8898	0.0744	0.8953	0.8823
	EDCSTFN	0.0280	1.0039	0.0891	0.8543	0.8833
	GAN-STFM	0.0232	0.7973	0.0707	0.8954	0.8944
	SwinSTFM	0.0223	0.8726	0.0712	0.9141	0.9030
	BPF-MGAN	0.0222	0.7608	0.0661	0.9090	0.9046
	CTUGAN	**0.0158**	**0.5314**	**0.0492**	**0.9525**	**0.9226**
	Ideal value	0	0	0	1	1

provides more stable and robust performance on diverse abrupt scenarios.

For LGC scenarios, there was a massive flooding in December 2004 and the land-cover type drastically changed. The Landsat and MODIS images of December 28, 2004, are taken as the prior data to predict the Landsat image on January 13, 2005. Fig. 12 depicts the predictions of various approaches for the LGC scenario of January 13, 2005. Fig. 12(a) shows the Landsat image on December 28, 2004; Fig. 12(b) shows the MODIS image on January 13, 2005; Fig. 12(c) shows the Landsat image on January 13, 2005; and Fig. 12(d)–(j) illustrates the predictions for Fig. 12(b) produced by the STARFM, FSDAF, EDCSTFN, GAN-STFM, SwinSTFM, BPF-MGAN, and CTUGAN approaches, respectively. The white and yellow boxes in Fig. 12 are enlarged and exhibited in Figs. 13 and 14. The MAEM between the prediction and ground truth is presented in Fig. 15.

In Fig. 12, the spectral information of the various predictions seems similar overall and the colors remain consistent with that of Fig. 12(c); however, the prediction of regions with the flooding greatly differs. It is evidently recognized that the STARFM, FSDAF, and EDCSTFN models poorly predict the land-cover changes after the flood, whereas the prediction effect of the CTUGAN is closest to Fig. 12(c).

From area 1 in Fig. 13, it is evident that the STARFM, FSDAF, EDCSTFN, GAN-STFM, SwinSTFM, and BPF-MGAN are poor in forecasting the land-cover changes after the flooding and depend on the reference data, while the prediction of the CTUGAN is superior. For region 2, the FSDAF, EDCSTFN, GAN-STFM, and SwinSTFM yield worse predictions, followed by the STARFM; compared to the CTUGAN, the BPF-MGAN preserves more structure, but more spectral information is lost. For region 3, the predictions of the CTUGAN are optimal for details, textures, edges, and spectra.

In Fig. 14, the CTUGAN produces more precise results than the BPF-MGAN for predicting the changes in the pink (region 1), white (region 2), and yellow (region 3) boxes. From regions 1 and 2 in Fig. 14, it is discovered that the STARFM, FSDAF, EDCSTFN, and GAN-STFM provide the worse predictions for significant changes, followed by the BPF-MGAN; the SwinSTFM and CTUGAN yield better results, while the CTUGAN result is closer to the reference data. For region 3 in Fig. 14, the STARFM, FSDAF, EDCSTFN, GAN-STFM, and SwinSTFM have the worse results; the BPF-MGAN and CTUGAN have better results, and the CTUGAN result comes closer to the ground truth. Table IV reveals that all the indexes of the CTUGAN are optimal. In Fig. 15, the MAEM of the CTUGAN is minimal. In summary, the STF result of the CTUGAN on the LGC data is optimal in both visual assessment and metrics and exhibits better spectral and structural fidelity.

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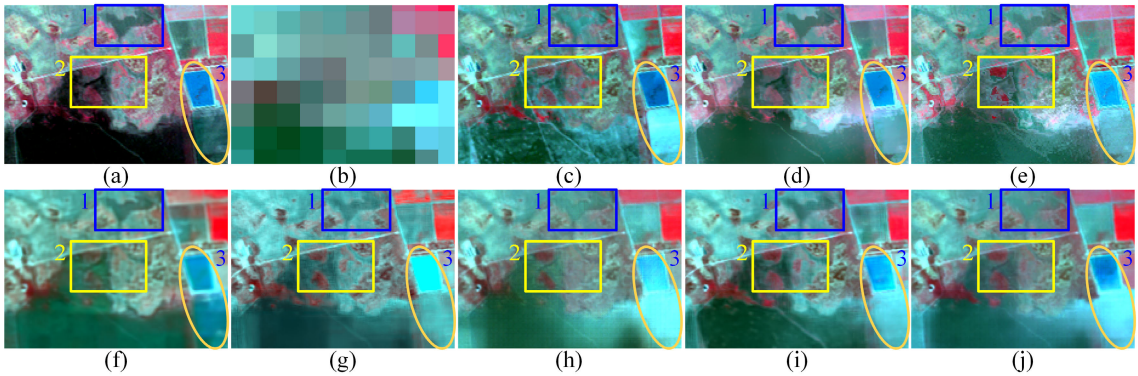


Fig. 13. Magnified contents of the white box in Fig. 12. (a) Landsat image on December 28, 2004. (b) MODIS image on January 13, 2005. (c) Landsat image on January 13, 2005, i.e., ground truth. (d) STARFM. (e) FSDAF. (f) EDCSTFN. (g) GAN-STFM. (h) SwinSTFM. (i) BPF-MGAN. (j) CTUGAN.

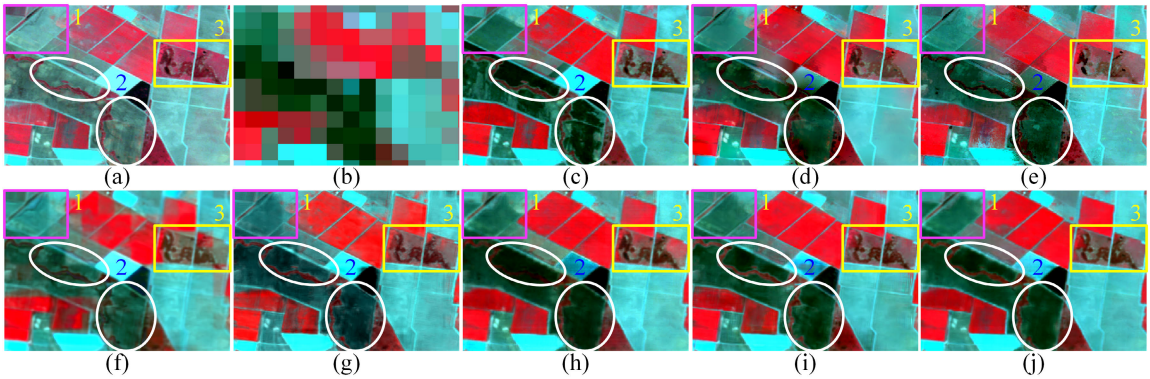


Fig. 14. Magnified contents of the yellow box in Fig. 12. (a) Landsat image on December 28, 2004. (b) MODIS image on January 13, 2005. (c) Landsat image on January 13, 2005, i.e., ground truth. (d) STARFM. (e) FSDAF. (f) EDCSTFN. (g) GAN-STFM. (h) SwinSTFM. (i) BPF-MGAN. (j) CTUGAN.

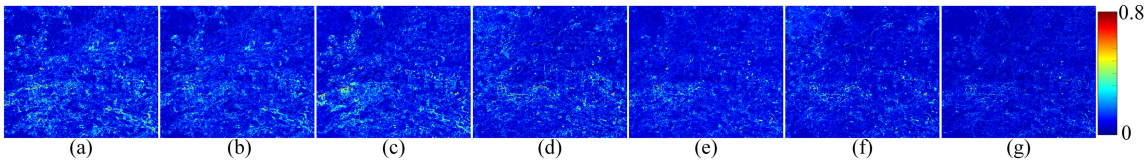


Fig. 15. MAEM between the prediction and ground truth. (a) STARFM. (b) FSDAF. (c) EDCSTFN. (d) GAN-STFM. (e) SwinSTFM. (f) BPF-MGAN. (g) CTUGAN.

E. Ablation Experiments

To validate the effects of the proposed RMDCT, MFF, MCAFT, MCTDs, and hyperparameters in the loss function on the fusion performance, ablation experiments are conducted on these components.

1) Validity of the RMDCT: The RMDCT is an essential component of the proposed CTUGAN for extracting local–global features. To prove the effectiveness of the RMDCT and the feasibility of this combination, sequences of ablation experiments are carried out. RMDCT consists of the residual MDconv and MDCT; therefore, three variants are devised: 1) remove the RMDCT, keeping the rest unchanged, recorded as without (w/o) RT; 2) remove the MDCT, having only the residual MDconv component and holding the rest unchanged, denoted as w/o CT; and 3) remove the residual MDconv and hold the rest unchanged, recorded as w/o MDC. The results of the three variants on the QB and CIA test data are displayed in Tables V and VI. From Tables V and VI, it is noticed that the metrics of w/o RT, w/o CT, and w/o MDC are worse than those of the proposed CTUGAN for both SSF and STF, and all metrics of the CTUGAN outcome are optimal. This demonstrates the positive effect of RMDCT.

TABLE V

MEAN VALUE OF INDEXES FOR ABLATION EXPERIMENTAL RESULTS OF THE RMDCT AND MFF ON THE QB TEST DATA

Data	Model	D_λ	D_S	QNR
QB	w/o RT	0.0226	0.0195	0.9584
	w/o CT	0.0192	0.0178	0.9634
	w/o MDC	0.0108	0.0122	0.9772
	w/o MFF	0.0157	0.0126	0.9719
	CTUGAN	**0.0047**	**0.0073**	**0.9880**

2) Validity of the MFF: The skip-connection components SF_n and SC_n $(n = 0, \ldots, N)$ utilized for fusion in CTUGAN are directly removed, and F_n and C_n are directly transferred to the MCAFT, holding the rest unchanged, recorded as w/o MFF. The experimental outcomes on the QB and CIA test data are given in Tables V and VI. It is noticed that both structural and spectral indexes of the w/o MFF are poorer than that of the proposed CTUGAN, demonstrating that the low-level multiscale features SF_n and SC_n $(n = 0, \ldots, N)$

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TABLE VI

INDEXES FOR ABLATION EXPERIMENTAL RESULTS OF THE RMDCT AND MFF ON THE CIA TEST DATA OF NOVEMBER 25, 2001

Data	Model	RMSE	ERGAS	SAM	CC	SSIM
CIA	w/o RT	0.0297	0.9695	0.1086	0.8248	0.8301
	w/o CT	0.0290	0.9442	0.1060	0.8359	0.8360
	w/o MDC	0.0273	0.8908	0.1011	0.8471	0.8431
	w/o MFF	0.0283	0.9151	0.1086	0.8439	0.8452
	CTUGAN	**0.0206**	**0.6640**	**0.0654**	**0.9169**	**0.8830**

TABLE VII

EFFECT OF THE MCAFT ON THE FUSION PERFORMANCE FOR THE QB DATA

Data	Model	D_λ	D_S	QNR
QB	MT-MLP	0.0104	0.0150	0.9747
	MT-C	0.0140	0.0166	0.9697
	MT-R	0.0103	0.0135	0.9763
	CTUGAN	**0.0047**	**0.0073**	**0.9880**

TABLE VIII

EFFECT OF THE MCAFT ON THE FUSION PERFORMANCE FOR THE CIA DATA OF NOVEMBER 25, 2001

Data	Model	RMSE	ERGAS	SAM	CC	SSIM
CIA	MT-MLP	0.0269	0.9079	0.1028	0.8652	0.8536
	MT-C	0.0276	0.8798	0.0991	0.8510	0.8520
	MT-R	0.0259	0.8546	0.0883	0.8667	0.8576
	MT-DA	0.0241	0.7794	0.0771	0.8945	0.8685
	MT-TA	0.0227	0.7308	0.0691	0.8979	0.8736
	CTUGAN	**0.0206**	**0.6640**	**0.0654**	**0.9169**	**0.8830**

supply effective structural details and spectral information during fusion.

3) Validity of the MCAFT: The MCAFT is a vital component for integrating information from diverse sources, which contains the MDCCA and MDCT. To demonstrate the effect of MDCCA in the fusion, two fully connected layers (MLP), convolution [Conv(k3)], and residual convolution are utilized to replace the MDCCA, and the other parts are kept invariant, denoted as MT-MLP, MT-C, and MT-R. Different modes of cross-attention generation are also validated on the CIA scenario. One case is that V is derived from the fusion result Y'_{n+1} at lower resolution; Q and K are derived from C'_n and F'_n, and the other parts are kept invariant, named MT-DA. The other case is that the cross-attention operation is performed on the dual-source information (C'_n and F'_n) and the fusion result Y'_{n+1}, respectively, and the two enhanced features are then concatenated and fused to generate T_n, and the other parts are kept invariant, denoted as MT-TA. The metrics for evaluating the outcomes on the QB and CIA test data are presented in Tables VII and VIII. Compared to the CTUGAN model, the performance of the MT-MLP, MT-C, and MT-R is significantly reduced for SSF and STF. From Table VIII, it is evident that the RMSE, ERGAS, SAM, CC, and SSIM metrics of the fusion results under MDCCA are superior to those of MT-DA and MT-TA. Thereby, it demonstrates the effectiveness of the cross-attention interaction mode of MDCCA.

4) Validity of the MCTDs: The MDCT plays a crucial role in MCTDs. To exhibit the effect of CTD and different resolution discriminators, ablation experiments are

TABLE IX

EFFECT OF THE MCTDS ON THE FUSION PERFORMANCE FOR THE QB DATA

Data	Model	D_λ	D_S	QNR
QB	MCD	0.0103	0.0171	0.9729
	MRD	0.0082	0.0159	0.9760
	CTD-1	0.0090	0.0129	0.9783
	CTD-2	0.0086	0.0128	0.9787
	CTD-3	0.0062	0.0090	0.9848
	CTD-4	0.0051	0.0084	0.9865
	CTUGAN	**0.0047**	**0.0073**	**0.9880**

TABLE X

EFFECT OF THE MCTDS ON THE FUSION PERFORMANCE FOR THE CIA DATA OF NOVEMBER 25, 2001

Data	Model	RMSE	ERGAS	SAM	CC	SSIM
CIA	MCD	0.0264	0.8902	0.0932	0.8695	0.8544
	MRD	0.0256	0.8381	0.0925	0.8717	0.8582
	CTD-1	0.0260	0.8552	0.0878	0.8649	0.8579
	CTD-2	0.0258	0.8096	0.0974	0.8789	0.8628
	CTD-3	0.0232	0.8292	0.0741	0.8997	0.8738
	CTD-4	0.0215	0.6940	0.0725	0.9153	0.8765
	CTUGAN	**0.0206**	**0.6640**	**0.0654**	**0.9169**	**0.8830**

implemented as follows: 1) replace the MDCT with the 3×3 convolution, denoted as MCD; 2) replace the MDCT with the 3×3 residual convolution, denoted as MRD; 3) with (w) CTD_0, recorded as CTD-1; 4) w CTD_0 and CTD_1, denoted by CTD-2; 5) w CTD_0, CTD_1, and CTD_2, written CTD-3; 6) w CTD_0, CTD_1, CTD_2, and CTD_3, denoted by CTD-4; and 7) w CTD_0, CTD_1, CTD_2, CTD_3, and CTD_4, denoted by CTD-5, i.e., CTUGAN. The objective assessment of results implemented on the QB and CIA test data is displayed in Tables IX and X. All the metrics of the MCD, MRD, and combination of different resolution discriminators are poorer than those of the CTUGAN, proving the positive effect of the MDCT and multiresolution discriminators in adversarial learning.

5) Hyperparameters in the Loss Function: To verify the influence of the hyperparameters in $\mathcal{L}_A$ on the fusion capacity, ablation experiments are conducted. Since both $\mathcal{L}_{FM}$ and $\mathcal{L}_{PL}$ are feature-level losses, which play an equally important role in $\mathcal{L}_A$, assign $\rho_2 = \varsigma_3$. $\mathcal{L}_{SaC}$ and $\mathcal{L}_{SeC}$ are the spatial detail consistency loss and spectral consistency (SeC) loss, which have equal importance; therefore, let $\lambda_4 = \xi_5$.

Experiments are performed on the CIA scenario of November 25, 2001. Based on empirical values, let $\mu_1 = 1 \times 10^{-3}$, $\rho_2 = \varsigma_3 = 1$, and $\lambda_4 = \xi_5 = 0.05$. Second, keep $\mu_1 = 1 \times 10^{-3}$ and $\lambda_4 = \xi_5 = 0.05$, and adjust the hyperparameters ρ_2 and ς_3 to 10^{-5}, 10^{-4}, 10^{-3}, 0.01, 0.1, 1, and 10 separately. The experimental results are provided in Table XI, and it is evidently observed that the fusion capability of the model is optimal at $\rho_2 = \varsigma_3 = 1 \times 10^{-3}$, so $\rho_2 = \varsigma_3 = 1 \times 10^{-3}$ is fixed. Third, keep $\mu_1 = 1 \times 10^{-3}$ and $\rho_2 = \varsigma_3 = 1 \times 10^{-3}$, and adjust the hyperparameters λ_4 and ξ_5 to 5×10^{-5}, 5×10^{-4}, 5×10^{-3}, 0.05, 0.5, 5, and 50 individually. The experimental results are provided in Table XI, and it is evidently observed that the fusion capability of the model is optimal at $\lambda_4 = \xi_5 = 5 \times 10^{-4}$, so $\rho_2 = \varsigma_3 = 1 \times 10^{-3}$ and $\lambda_4 = \xi_5 = 5 \times 10^{-4}$ are fixed. Fourth, keep $\rho_2 = \varsigma_3 = 1 \times 10^{-3}$ and $\lambda_4 = \xi_5 = 5 \times 10^{-4}$,

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TABLE XI
EFFECT OF THE HYPERPARAMETERS OF THE LOSS FUNCTION ON THE MODEL PERFORMANCE

Data	Setting			RMSE	ERGAS	SAM	CC	SSIM
CIA	CTUGAN			**0.0206**	**0.6640**	**0.0654**	**0.9169**	**0.8830**
	$\mu_1 = 1 \times 10^{-3}$ $\lambda_4 = \xi_5 = 0.05$	$\rho_2 = \varsigma_3$	10^{-5}	0.0220	0.7301	0.0757	0.9109	0.8733
			10^{-4}	0.0215	0.7433	0.0735	0.9075	0.8799
			10^{-3}	0.0212	0.7127	0.0696	0.9083	0.8808
			0.01	0.0213	0.7251	0.0713	0.9085	0.8803
			0.1	0.0218	0.7285	0.0722	0.9019	0.8756
			1	0.0216	0.7433	0.0711	0.9059	0.8777
			10	0.0224	0.7947	0.0735	0.9003	0.8754
	$\mu_1 = 1 \times 10^{-3}$ $\rho_2 = \varsigma_3 = 1 \times 10^{-3}$	$\lambda_4 = \xi_5$	5×10^{-5}	0.0209	0.7103	0.0706	0.9120	0.8829
			5×10^{-4}	**0.0206**	**0.6640**	**0.0654**	**0.9169**	**0.8830**
			5×10^{-3}	**0.0206**	0.6728	0.0679	0.9157	0.8780
			0.05	0.0212	0.7127	0.0696	0.9083	0.8808
			0.5	0.0214	0.7466	0.0724	0.9081	0.8802
			5	0.0217	0.7797	0.0715	0.9061	0.8785
			50	0.0216	0.7040	0.0695	0.9050	0.8709
	$\lambda_4 = \xi_5 = 5 \times 10^{-4}$ $\rho_2 = \varsigma_3 = 1 \times 10^{-3}$	μ_1	10^{-5}	0.0225	0.7611	0.0758	0.9051	0.8778
			10^{-4}	0.0211	0.7267	0.0700	0.9093	0.8811
			10^{-3}	**0.0206**	**0.6640**	**0.0654**	**0.9169**	**0.8830**
			0.01	0.0213	0.7261	0.0711	0.9083	0.8803
			0.1	0.0217	0.7450	0.0747	0.9058	0.8789
			1	0.0216	0.7073	0.0756	0.9113	0.8737

TABLE XII
EFFECT OF THE COMPONENTS OF THE LOSS FUNCTION ON THE MODEL PERFORMANCE

Data	Setting					RMSE	ERGAS	SAM	CC	SSIM
	$\mathcal{L}_{\text{ACTG}}^{\text{RaLS}}$	$\sum\mathcal{L}_{\text{FM}}$	$\mathcal{L}_{\text{PL}}$	$\mathcal{L}_{\text{SaC}}$	$\mathcal{L}_{\text{SeC}}$					
CIA	√	√	√	√	√	**0.0206**	**0.6640**	**0.0654**	**0.9169**	**0.8830**
		√	√	√	√	0.0211	0.6873	0.0691	0.9085	0.8726
	√		√	√	√	0.0221	0.7271	0.0719	0.9037	0.8760
	√	√		√	√	0.0218	0.7703	0.0771	0.9072	0.8788
	√	√	√		√	0.0233	0.7827	0.0784	0.8958	0.8718
	√	√	√	√		0.0230	0.9024	0.0790	0.9049	0.8738

and adjust the hyperparameter μ_1 to 10^{-5}, 10^{-4}, 10^{-3}, 0.01, 0.1, and 1. The experimental results are provided in Table XI, and it is evidently observed that the fusion capability of the model is optimal at $\mu_1 = 1 \times 10^{-3}$, so $\rho_2 = \varsigma_3 = 1 \times 10^{-3}$, $\lambda_4 = \xi_5 = 5 \times 10^{-4}$, and $\mu_1 = 1 \times 10^{-3}$ are fixed. From the experimental results, it is noticed that the variation of hyperparameters does not have too much impact on the fusion performance.

Furthermore, the effect of the lack of one of the five losses is verified on the CIA test data, and the experimental results are displayed in Table XII. The metrics are all worse when the multiresolution adversarial loss is excluded. When FM loss or PL is absent, the indicators are worse, and the introduction of FM loss or PL enhances the image quality. The metrics significantly worsen when the spatially consistent loss or spectrally consistent loss is removed.

F. Model Complexity

The model complexity of the proposed CTUGAN approach and the comparative methods are evaluated in terms of parameters and floating point operations (FLOPs). For SSF, the FLOPs for CTUGAN are obtained on the inputs with dimensions of $256 \times 256 \times 1$ and $16 \times 16 \times 4$. For STF, the FLOPs for CTUGAN are obtained on the inputs with dimensions of $256 \times 256 \times 6$ and $16 \times 16 \times 6$. For other comparative models, the size of the high-spatial-resolution input is also 256×256, and the channel number is set according to the original article.

The model parameters (M: 10^6) and FLOPs (G: 10^9) obtained from the SSF on GF-2 data and the STF on LGC data are presented in Tables XIII and XIV. From Table XIII, it is evidently observed that the proposed CTUGAN has the most parameters, whereas the downsampling operation of CTUGAN leading to the decreased size of the data and the large number of dense connections in the BPF-MGAN make the FLOPs of CTUGAN lie in the middle among the compared methods. From Table XIV, it is noticeable that the number of parameters in CTUGAN is slightly less than that of SwinSTFM.

IV. DISCUSSION

The CTUGAN tackles the lack of DL models capable of simultaneously performing SSF and STF via the BALGE,

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TABLE XIII
MODEL COMPLEXITY ON GF-2 DATA

Data	Model	Parameters (M)	FLOPs (G)
GF-2	RED-cGAN	1.90	75.54
	PsGAN	1.89	48.40
	DIGAN	0.13	31.65
	PGMAN	0.39	2.61
	UCGAN	0.21	16.04
	PanFormer	1.53	11.78
	BPF-MGAN	2.93	109.47
	CTUGAN	21.43	41.97

TABLE XIV
MODEL COMPLEXITY ON LGC DATA

Model	EDCSTFN	GAN-STFM	SwinSTFM	BPF-MGAN	CTUGAN
Parameters (M)	0.28	0.58	39.67	2.95	21.46
FLOPs (G)	18.59	37.76	28.19	109.76	42.05

MCAFD, and MCTDs, which provides superior fusion performance to mainstream SSF and STF methods.

For the SSF experimental results, compared with supervised and unsupervised SSF approaches, the CTUGAN yields better results on real-world data. From the full-resolution experimental results on GF-2 and QB, it is observed that the conventional approaches tend to suffer from spectral distortions and produce low-quality spatial structures. The supervised methods are unable to properly transfer the knowledge obtained from the reduced-resolution model to the full-resolution scenarios, which appear to have more spectral distortions, image blurring, and artifacts. Unsupervised approaches yield superior outcomes on full-resolution scenes owing to the knowledge learned directly from the source images, whereas the proposed CTUGAN algorithm combines supervised and unsupervised strategies to fully learn the prior information in the source images, which preserves more spectral information and spatial structure and produces superior-quality images. Furthermore, the CTUGAN is remarkably superior in metrics of spectral, spatial structure, and global quality assessment. The above demonstrates the outstanding capability of the proposed CTUGAN to generate HRMS images that meet the requirements of remote sensing missions in real-world scenarios.

For STF experimental results, though the CTUGAN only exploits two images, the fusion effect is more precise than the models employing three images, which promotes the facility for practical application. Moreover, the proposed CTUGAN yields superior STF results on various CIA and LGC test scenarios. For example, from the analysis of indicators in Tables III and IV, for traditional methods, the ability of FSDAF to predict phenological changes and land-cover changes is generally superior to that of STARFM. Even the prediction of massive floods on December 12, 2004, is more precise than the CNN-based models of EDCSTFN and GAN-STFM. The FSDAF is more reliable than the EDCSTFN and the GAN-STFM in predicting phenological changes in the first, second, and fourth groups of CIA data, whereas the STF capability of the EDCSTFN and GAN-STFM differs on different CIA scenes. The STF ability of the EDCSTFN outperforms that of the GAN-STFM on the first and second groups of CIA scenes, and the STF ability of the GAN-STFM is superior to that of the EDCSTFN on the prediction of phenological changes on the third, fourth, and fifth groups of CIA scenes. SwinSTFM benefits from the establishment of long-range dependencies and has superior STF capability over the EDCSTFN and GAN-STFM. The CTUGAN performance outperforms our previous work, BPF-MGAN, in both SSF and STF, which benefits from the superiority of the convolutional Transformer structure for local and global feature representation. The convolution is limited for the extraction of long-range dependency, and expanding the receptive field takes massive convolutions; therefore, the tradeoff between model complexity and fusion power makes the BPF-MGAN inferior to the CTUGAN in capturing long-range dependencies. From the analysis of the visual evaluation, the proposed CTUGAN prediction of phenological changes and land-cover-type changes is closer to the reference image and is least influenced by the prior data.

The CTUGAN can provide superior SSF capability to realize the fusion of a low-spatial-resolution MS image with a low-spectral-resolution PAN image to generate the HRMS image. In addition, the CTUGAN can also provide superior STF capability to realize the fusion of a time-dense low-spatial-resolution MS image with a time-sparse high-spatial-resolution MS image to generate the HTHR MS image. The CTUGAN overcomes the mutual limitations of a single-source image in temporal, spatial, and spectral resolution, which is of vital theoretical significance and application value to enhance the utility of the image and the ability of Earth observation. The CTUGAN can provide both reliable and high-quality data for remote sensing applications and provide technical and theoretical support. The CTUGAN can produce high-quality HRMS images, which are significant in improving the reliability of applications [75], [76] such as resource exploration, land-cover/utilization classification [77], [78], target detection and recognition (aircraft, ship, vehicle, airport, port, building, and so on) [79], [80], crop mapping [81], and so on. The CTUGAN may also attempt to implement the application in the declouding of RSIs through fusion [82]. The CTUGAN provides high-quality HTHR MS images, which are of great significance in increasing the reliability of applications for dynamic monitoring and change detection of resources, for instance, disaster monitoring (drought monitoring, fire monitoring, flood monitoring, landslides, and so forth) [83], [84], [85], [86], atmospheric monitoring [87], vegetation monitoring (vegetation cover monitoring, deforestation and expansion, vegetation health monitoring, and so forth) [88], [89], [90], hydrological environment monitoring (coastal zone ecological environment monitoring, coastline extraction, wetland monitoring, and so forth) [91], [92], [93], urban construction [94], and crops (crop cultivation distribution maps, crop condition monitoring, crop yield estimation, crop disaster assessment, and so forth) [95], [96], [97].

Though the CTUGAN achieves superior performance in both SSF and STF, the CTUGAN has the following limitations. The CTUGAN predicts temporal variations only using the two images, lacking prior spatial and temporal variations. The

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CTUGAN is a data-driven model that is trained independently on four datasets of SSF and STF, instead of a single model applied to four datasets. The conjunction of RSI knowledge with data-driven models is promising to solve this problem. Future research will consider the integration of relevant knowledge with the data-driven model to further generalize the model and reduce model parameters. Heterogeneous fusion of radar and optical RSIs, and fusion of hyperspectral and MS images will be attempted in the future as well.

V. Conclusion

This article proposes a CTUGAN for SSF and STF. The CTUGAN contains an ACTG and MCTDs, both with the convolution and Transformer. In the ACTG, a BALGE is devised to extract local–global features via high-to-low resolution and low-to-high resolution. The MCAFD is devised to aggregate and fuse complementary local–global features of various hierarchies and resolutions for restoring valuable information. Moreover, the MCTDs adversely learn multiresolution local–global features to identify the relative reality of products, and a generalized loss function is built to accomplish full supervision. Finally, numerous experiments on the SSF and STF data demonstrate that the proposed CTUGAN outperforms in both subjective and objective evaluations.

Given the limitations of the above discussion, further exploration of data-knowledge joint drive, heterogeneous fusion, and hyperspectral fusion will be conducted in the future.

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